

Global Agricultural Outlook

Agricultural Production 1991-2019



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Acknowledgement

*I want to express my sincere gratitude to **Annalisa Occhipinti**, for the support and guidance throughout the creation of this report and Power BI project. Your expertise and insight were invaluable, and I couldn't have done it without your help.*

Your encouragement and belief in my abilities pushed me to put in my best effort and deliver a high-quality project. Your dedication to helping me succeed is greatly appreciated, and I am so grateful to have had the opportunity to work with you.

Thank you for everything you have done to help me bring this project to fruition. Your support has been invaluable, and I will always be grateful for your kindness and guidance.

Fabian Umeh

Executive Summary

This report examines the history of agricultural crop production, total output, and the role of agriculture in the global economy.

It shows that a significant number of countries do not produce many agricultural crops, including cereals, fruits, legumes, and roots. India is the leading producer of beans and cashews, while China is the top producer of apples, rice, wheat, potatoes, and sweet potatoes. Nigeria is the largest producer of yams and cassava, Russia is the second-largest producer of barley and rye, and the United States is the largest producer of soybeans and maize.



Globally,

- ◆ The root and tuber crop with the highest production, the potato, accounts for half of the total production.
- ◆ The most widely grown cereal crop is maize.
- ◆ A startling 85% of all production of legumes and nuts is made up of soybeans.
- ◆ The fruit that is produced the most worldwide is the banana.

Percentage change in Europe



Percentage change in Oceania



Globally,

- ◆ Among other continents Oceania and Europe has recorded the highest number of declines in percentage of agricultural output over the years.

1. Introduction

To fulfil the second Sustainable Development Goal of ending hunger, agricultural output must remain central. But while there are many factors to consider in developing a strong agricultural framework, sufficient insight always lies at its core. A thorough scrutiny is to be performed on historical data to identify any pattern trends or information that will be useful in facilitating future plans.

1.1 Data Source:

This analysis explores major areas of global agricultural trends from 1991 – 2019 with real datasets gotten from OWID (Our World In Data), it contains real world data of global agricultural historical statistics from (1991-2019). The major downside of this dataset was that all the information were not accumulated in a single table but rather were separated into different tables according to subject topics. However, this analysis made use of the different tables individually and merged some were necessary. Please note that a pre-processing step was performed using excel to make sure that the year considered for the analysis were consistent (1991-2019) before introducing the tables to Power BI.

Name	Description	Link
number-of-people-employed-in-ag	“Agriculture includes the cultivation of crops and livestock production, as well as forestry, hunting, and fishing. Employment includes anyone engaged in any activity to produce goods or services for pay or profit.”	“ https://ourworldindata.org/grapher/number-of-people-employed-in-agriculture ”
value of agricultural produce	“Gross production value of the agricultural sector, measured in current US\$”	“ https://ourworldindata.org/grapher/value-of-agricultural-production ”
Agricultural output	“Total agricultural output is the sum of crop and livestock products. It is measured in constant 2015 US\$, which means it adjusts for inflation.”	“ https://ourworldindata.org/grapher/agricultural-output-dollars?country=~OWID_WRL ”
National GDP	“National Gross Domestic Product. Adjusted for inflation and for differences in the cost of living between countries.”	“ https://ourworldindata.org/grapher/national-gdp ”
casava (root and tuber)	“Cassava production measured in tonnes.”	“ https://ourworldindata.org/grapher/cassava-production ”
yam (root and tuber)	“Yam production measured in tonnes.”	“ https://ourworldindata.org/grapher/yams-production ”
potato (root and tuber)	“Potato production measured in tonnes.”	“ https://ourworldindata.org/grapher/potato-production ”
sweet potato (root and tuber)	“Sweet potato production measured in tonnes.”	“ https://ourworldindata.org/grapher/sweet-potato-production ”
wheat (cereal)	“Wheat production measured in tonnes.”	“ https://ourworldindata.org/grapher/wheat-production ”
maize (cereal)	“Corn (maize) production is measured in tonnes.”	“ https://ourworldindata.org/grapher/maize-production ”
rice (cereal)	“Rice production measured in tonnes.”	“ https://ourworldindata.org/grapher/rice-production ”

Barley (cereal)	"Barley production measured in tonnes."	"https://ourworldindata.org/grapher/barley-production"
Rye (cereal)	"Rye production measured in tonnes."	"https://ourworldindata.org/grapher/rye-production"
Soybeans (legumes and nuts)	"Soybean production measured in tonnes."	"https://ourworldindata.org/grapher/soybean-production"
Beans (legumes and nuts)	"Bean production measured in tonnes."	"https://ourworldindata.org/grapher/bean-production"
Cashew (legumes and nuts)	"Cashew nut production measured in tonnes."	"https://ourworldindata.org/grapher/cashew-nut-production"
peas (Legumes and nuts)	"Pea production measured I tonnes."	"https://ourworldindata.org/grapher/pea-production"
Bananas (Fruits)	"Banana production measured in tonnes."	"https://ourworldindata.org/grapher/banana-production"
Orange (Fruits)	"Orange production measured I tonnes."	"https://ourworldindata.org/grapher/orange-production"
Apple (Fruits)	"Apple production measured in tonnes."	"https://ourworldindata.org/grapher/apple-production"
Grapes (Fruits)	"Grape production in tonnes."	"https://ourworldindata.org/grapher/grapes-production"
Avocado (Fruits)	"Avocado production measured in tonnes."	"https://ourworldindata.org/grapher/avocado-production"
GNI per capita	"GNI per capita is the ratio of gross national income and by mid-year population"	"http://data.un.org/Data.aspx?d=WDI&f=Indicator_Code%3ANY.GNP.PCAP.PP.CD"

1.2 Key Analysis:

This analysis explores major agricultural trends from 1991 – 2019, While providing insights into some important questions such as:

1. Which country or countries are the leading and least producers of food categories?

Proper addressing of this question accurately provides access to even more critical and effective questions such as where does the focus shifts to next? what are the underlying reasons behind the increase or decrease in food production of a country? what can be done to drive food production in least producing countries?

2. What is the agricultural value of a country and it's share in GDP?

Addressing this question equally provides a foundation to more advanced questions such as in which country is agriculture adding value? what can be done to improve the value added to a country's economy by agriculture?

3. What is the percentage change in agricultural output through the year 1991 – 2019?
4. By what percentage (larger or smaller) did the agricultural output differ from the year of reference (1991) and which country or countries has made the most significant change?
5. Which countries have managed to decouple agricultural output from more input?
6. What is the total agricultural value, total agricultural output and ranking of countries in terms of these factors?
7. Which crop type is the most produced continentally?

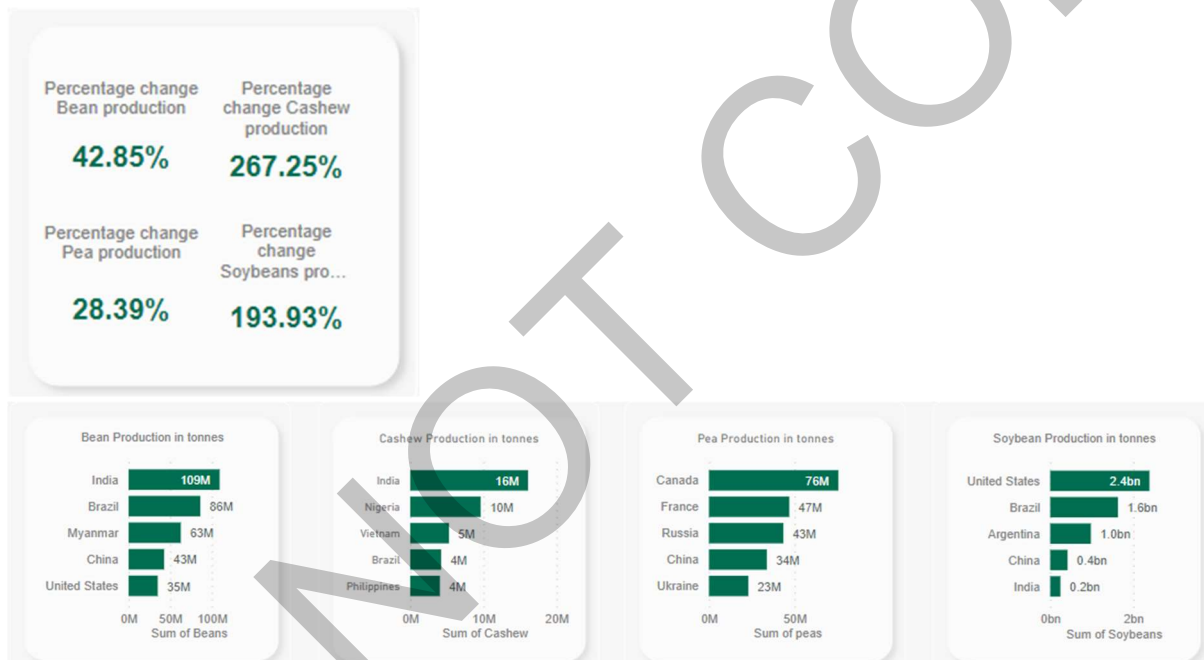
8. What percentage of agricultural market and global economy is held by the different continents?
9. How does agricultural output and year impact agricultural gross product value?
10. What is the Latest Income category of a country as per World Bank standards?

1. Which country or countries are the leading and least producers of food and how do they compare continentally and globally?

a) LEGUMES AND NUT ANALYSIS

- i. Which country or countries are the top and least producers of Legumes and nuts (1991-2019) and what are the percentage changes in production?

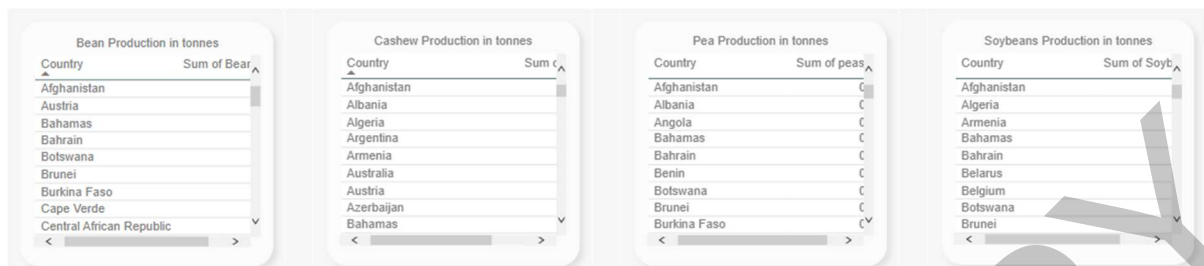
Top 5 producers of legumes and nuts



This chart shows the top 5 producers of legumes and nuts, and the percentage change in production. The percentage change can be selected for different year ranges, but the chart above shows the percentage change between 1991 and 2019 (default).

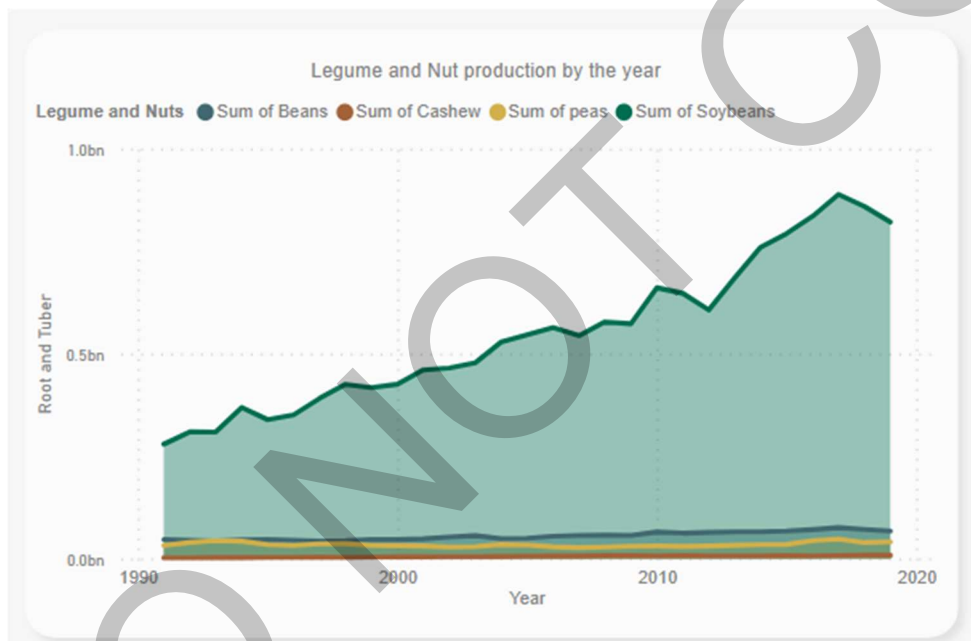
- India is the top producer of beans and cassava. India has produced 110 million tonnes of beans, about 25 million tonnes more than Brazil the second top producers and has produced 16 million tonnes of cashew nut almost 100 % the total production of Nigeria.
- The top producers of peas and soybeans are Canada and the United States respectively.
- The production of cashew and soybeans has increased by more than 100 percent in 2019 since 1991 reaching a total of 267.25% and 193.93% respectively. Bean and pea have shown significant increases too.

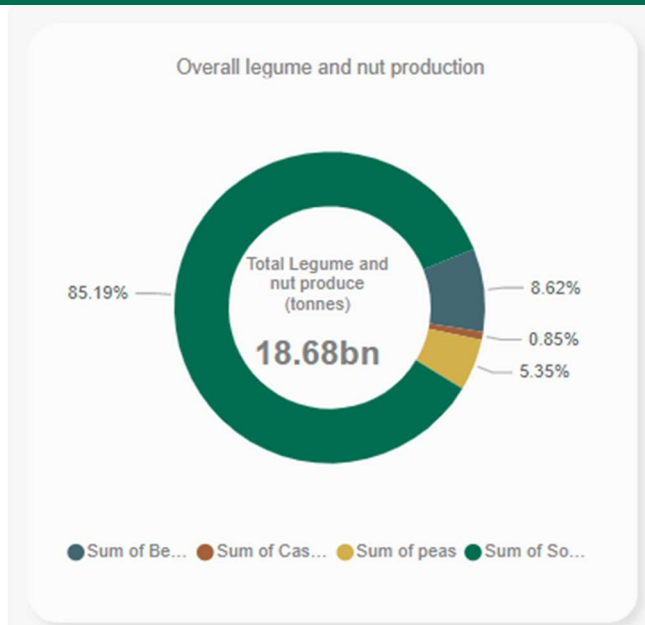
Least producers of legumes and nuts



The chart above shows the least producers or non-producers of legumes and nuts. There maybe little production in the countries above, but according to the data collected no significant production were recorded for such countries.

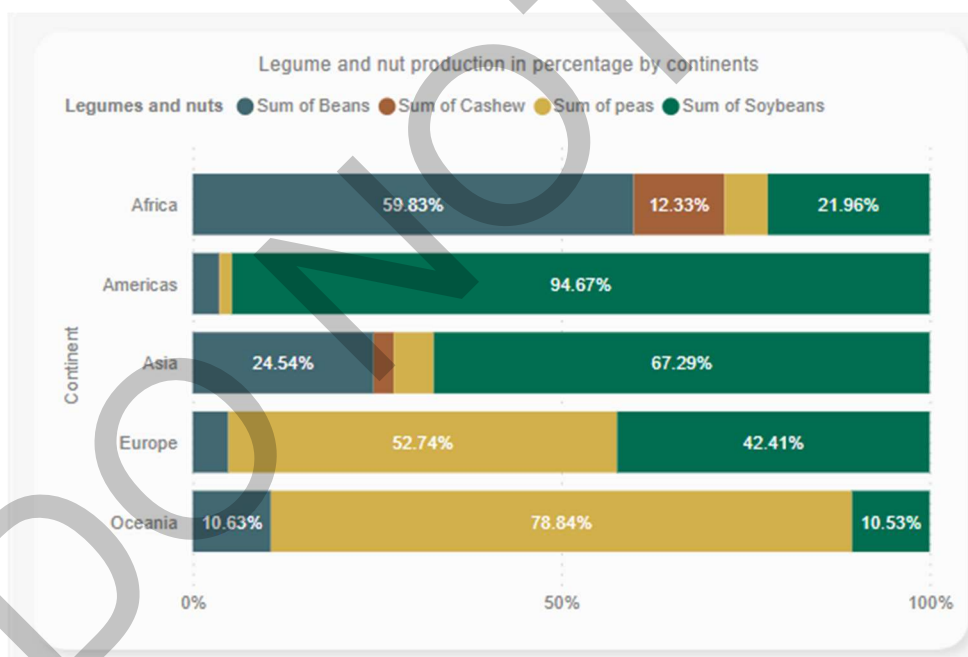
ii. Legumes and nuts production globally by year





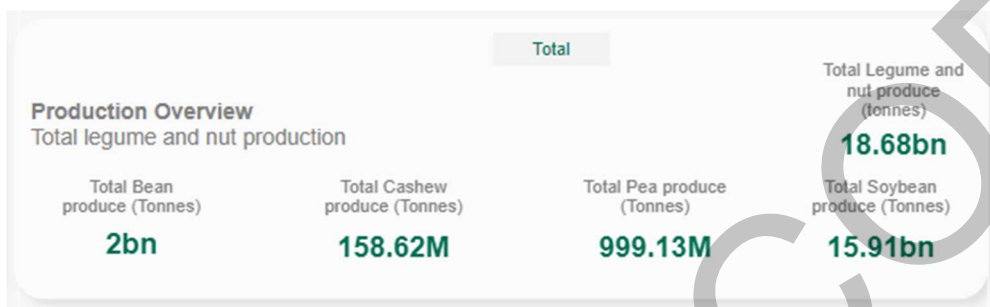
The chart above compares the production of the various legume and nut produces across the year 1991 to 2019. It reveals that soybeans among other legume and nut crops are the most produced globally in tonnes.

iii. Continent analysis on Legumes and nuts production



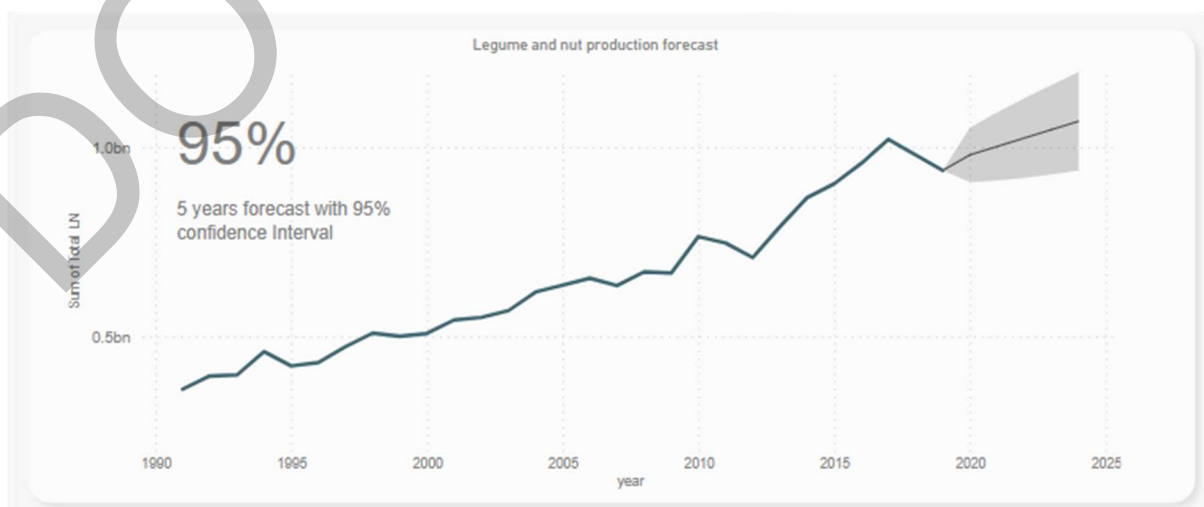
The chart above presents the production distribution in percentage of various legume and nut categories across the different continents of the world. In America and Asia, more than 70% of the total legume and nut produce comes from soybeans, while in Europe and Oceania, more than 50% of the total produce comes from peas. More than 50% of total Africa's Legume and nut produce comes from beans.

iv. What is the total and average legume and nut production (tonnes) and the rank of a country in terms of this produce?



The card reveals the total legume and nut production globally from 1991- 2019, upon using the year filter, it also reveals the total produce for the selected year or year range. The two cards by the right reveals the continental and global ranking of any selected country.

v. What will the estimated legume and nut production be in five years?



The line chart shows the total legume and nut produce from 1991 to 2019 globally and the forecast of future production with a 95% confidence interval. There was a dip in legume and nut produce in the year 2019. However, the forecast reveals growth in production for the next five years.

Filters used

Page Filters

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Country
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Year
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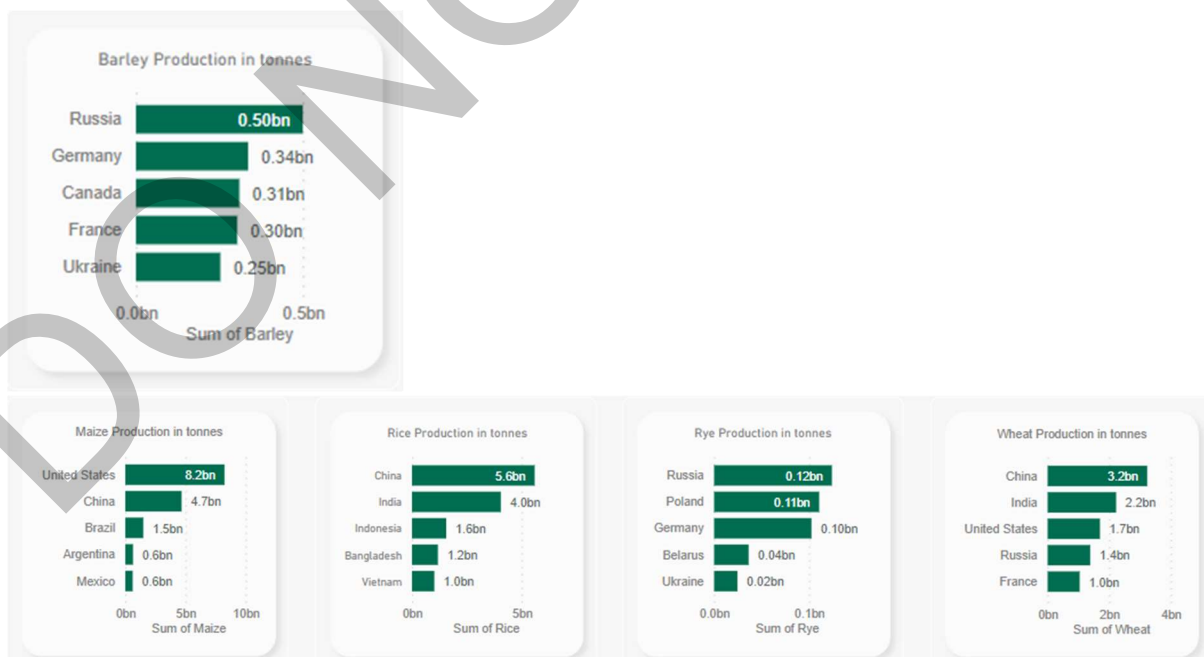
Top Bottom

The Top and Bottom buttons above enable the switch (ctrl + click) between the top and least producers of the different legume and nut crops, the filters give the user the leverage to select any country, continent and year or year range (hold ctrl) to analyse.

b) CEREAL ANALYSIS

- i. Which country or countries are the top and least producers of Cereal (1991-2019)?

Top 5 producers of Cereal crops



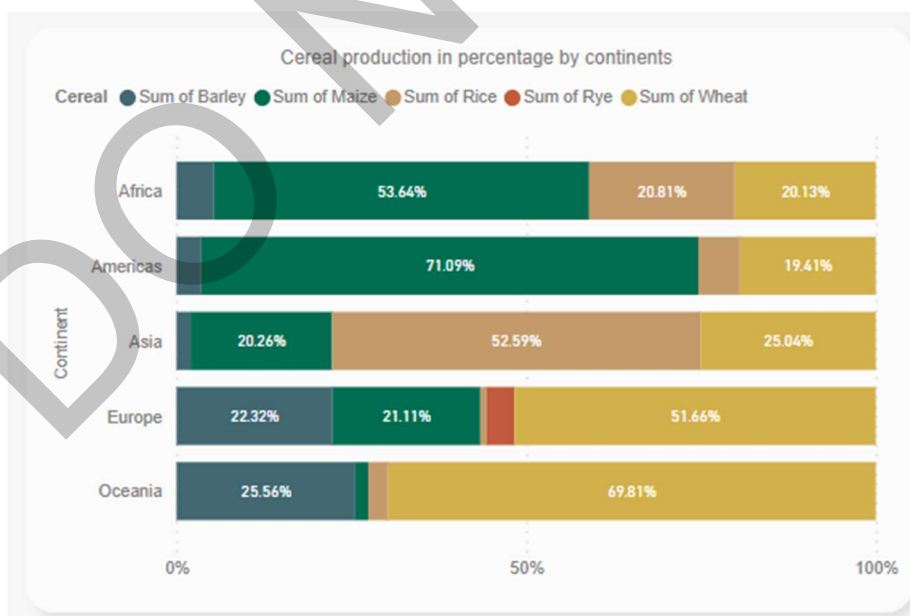
The chart above shows the top 5 producers of the different cereal crops globally. The top producers of wheat and rice globally is China, while Russia produces the largest amount of rye and barley globally. The United States is the top producers of Maize.

Least producers of cereal crops



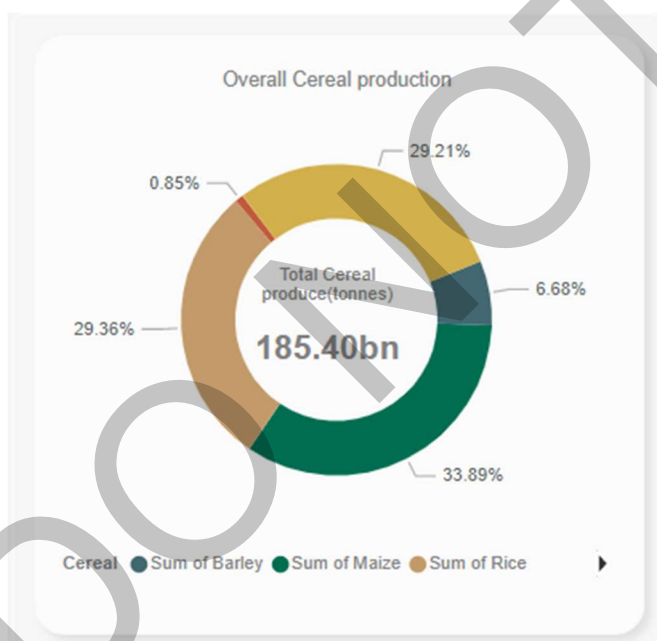
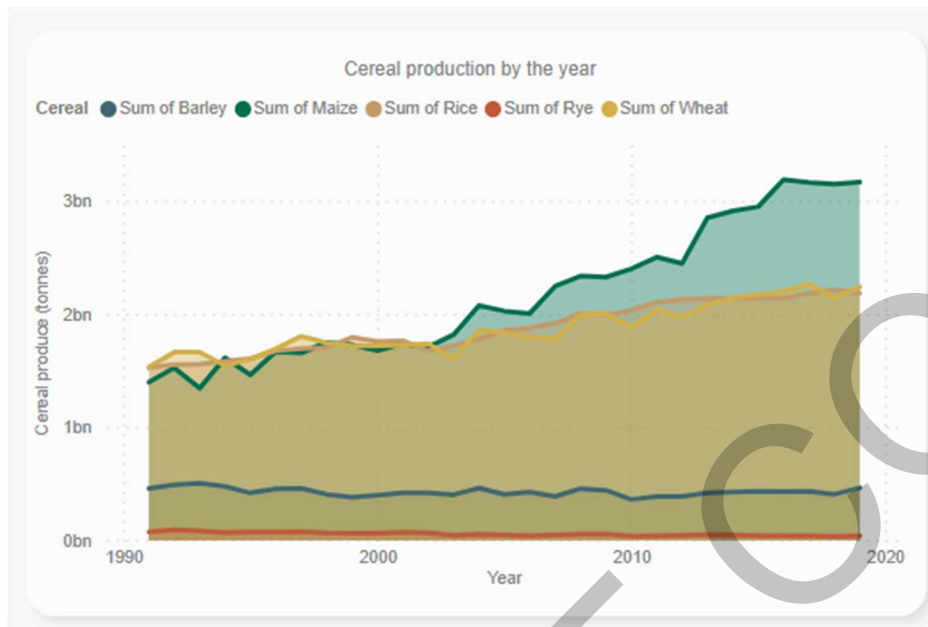
The table above shows the least producers of cereal. The data shows no significant production was recorded for such countries.

ii. What is the cereal production distribution like across the different continents?



The chart above shows cereal production distribution in percentage across the different continents of the world. Almost 70% of Oceania's cereal produce comes from wheat, more than 70% of Americas cereal produce comes from maize, about 50% of Europe's, Africa's and Asia's cereal produce comes from wheat, maize, and rice respectively.

iii. Global cereal production through the years



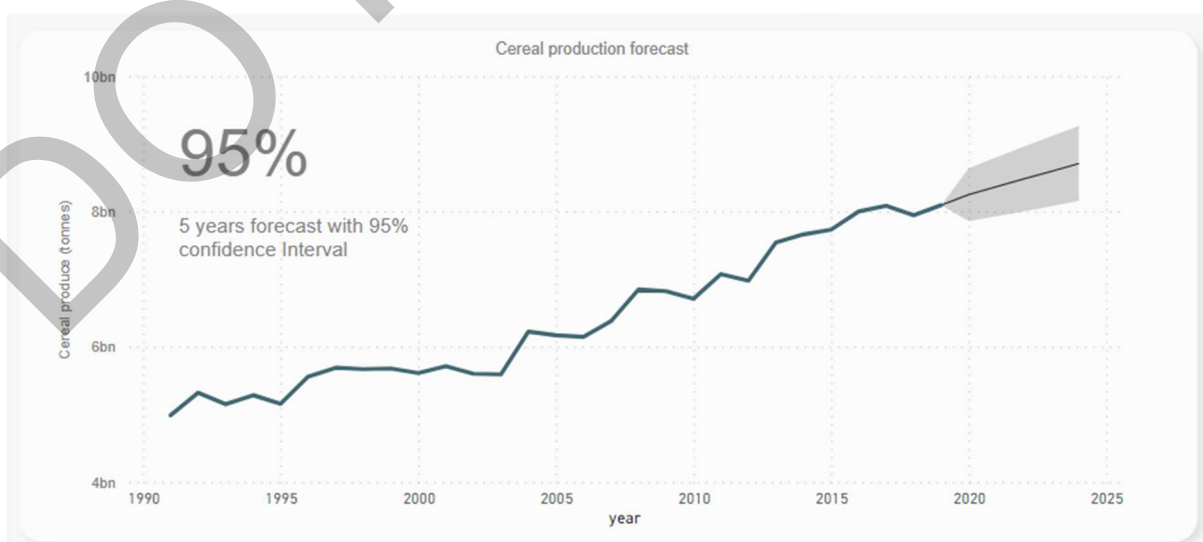
The area chart above shows the global cereal production across the year 1991 to 2019. Maize, rice, and wheat leads the chart in global cereal production, while barley and rye are the least produced globally.

iv. What is the total and average cereal production (tonnes) and the rank of a country in terms of this produce?



The card reveals the total cereal production globally from 1991- 2019, upon using the year filter, it also reveals the total produce for the selected year or year range. The two cards by the left reveals the continental and global ranking of any selected country.

- v. What will the estimated cereal production be in five years?



The line chart reveals that global cereal production will continue to rise and will exceed 8.5 billion before the year 2024 with a 95% confidence interval.

Filters used

Page Filters

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Country
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Year
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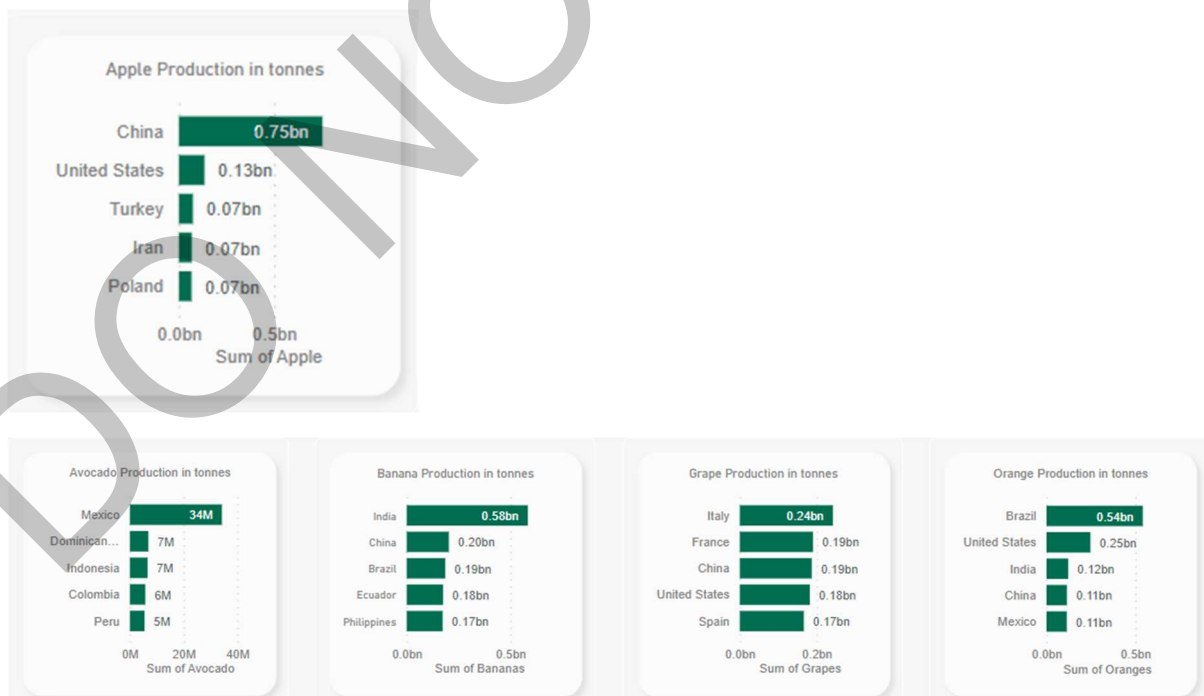
Top Bottom

The Top and Bottom buttons above enable the switch (ctrl + click) between the top and least producers of the different cereal crops, the filters give the user the leverage to select any country, continent and year or year range (hold ctrl) to analyse.

c) FRUITS ANALYSIS

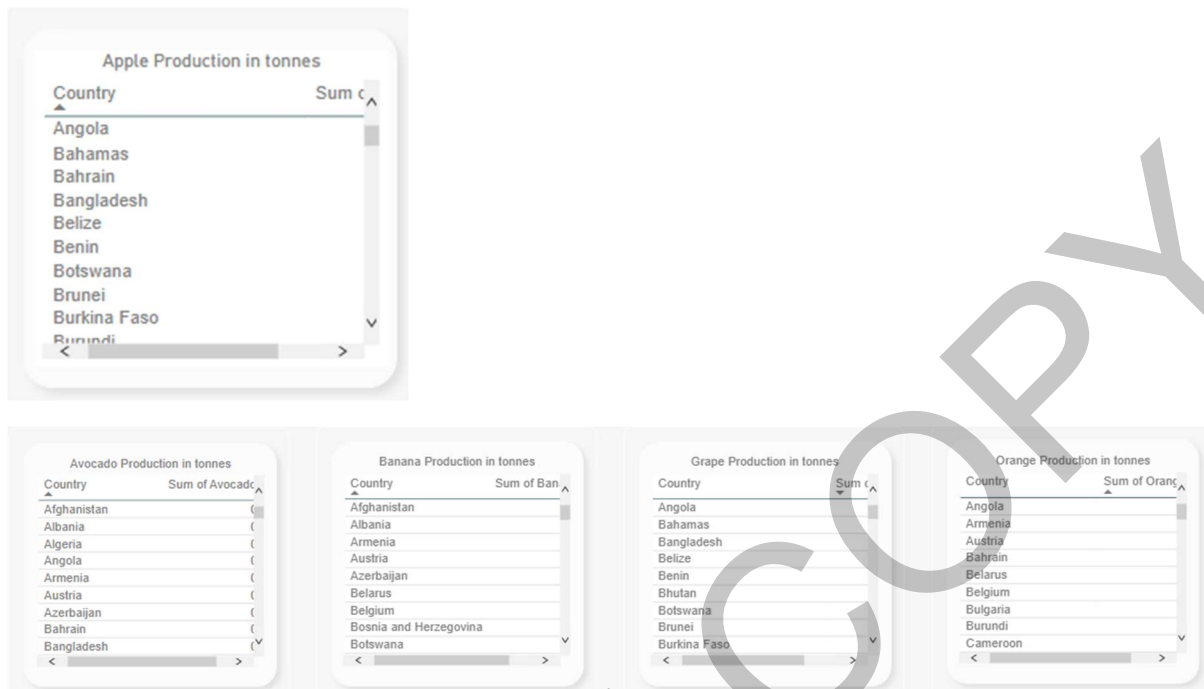
- i. Which country or countries are the top and least producers of fruits?

Top 5 producers of fruits

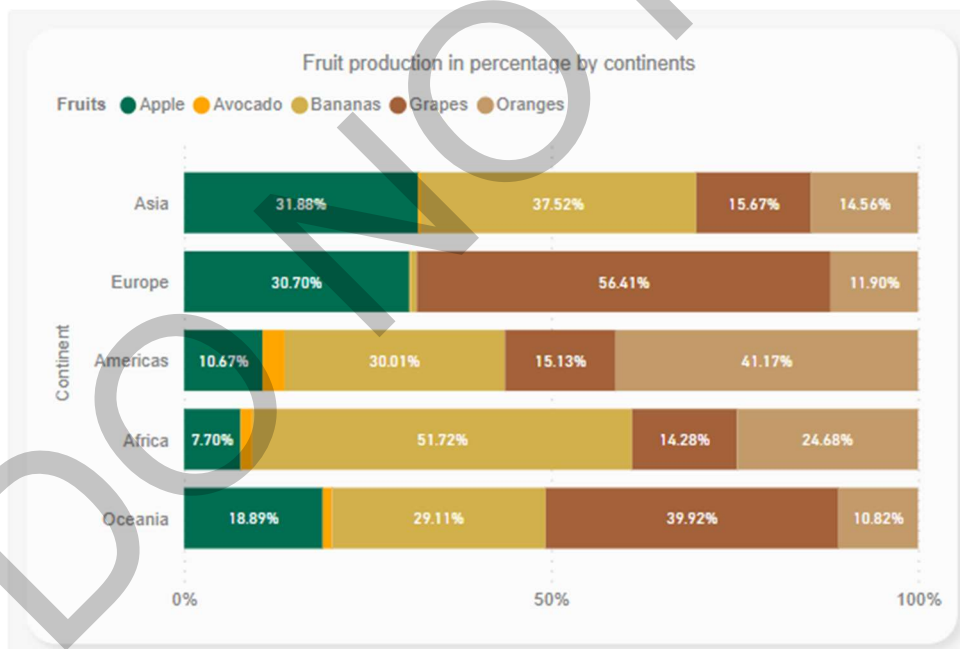


The chart above presents a suitable way to visualise the fruit production of the different countries of the world, the top 5 producers of each fruit types and the least producers of fruits and vegetables.

Least Producers of Fruits.

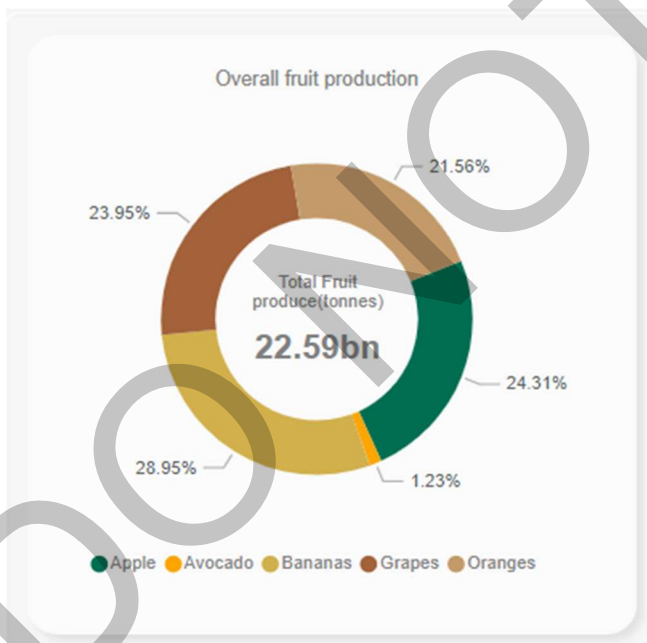
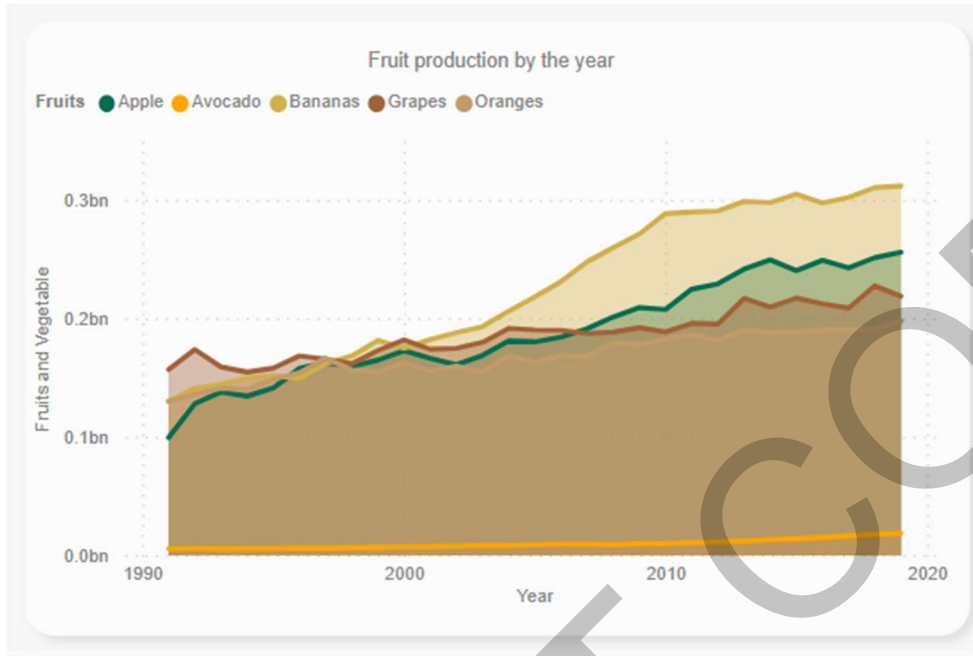


ii. What is the distribution of fruit produce in percentage among continents?



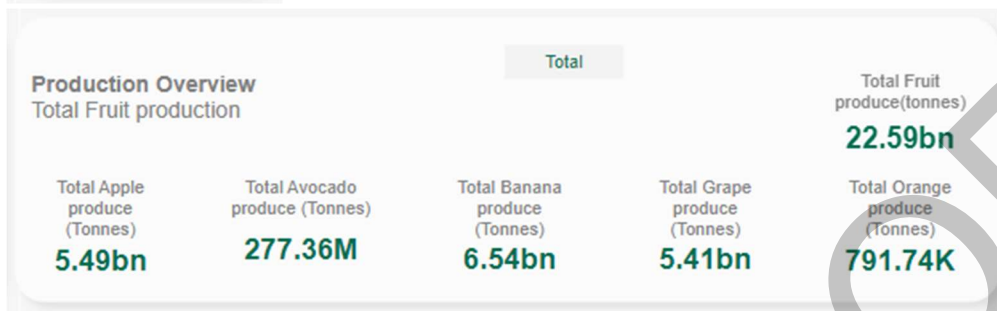
This chart creates a quick summary of the distribution in percentage of fruit production by continents from 1991-2019. Tomato is the major vegetable produced in Asia, Europe's grape production accounts for more than one third of the total fruit produce, Orange is the major fruit grown in the Americas, Africa produces more tomato compared to other fruit types while grape production in Oceania accounts for nearly half of the total fruit and vegetable produce. More so, if a particular country is selected, this chart also shows the percentage of fruit produced in that country.

- iii. Which fruit type are the most produced and least produced globally through the years?



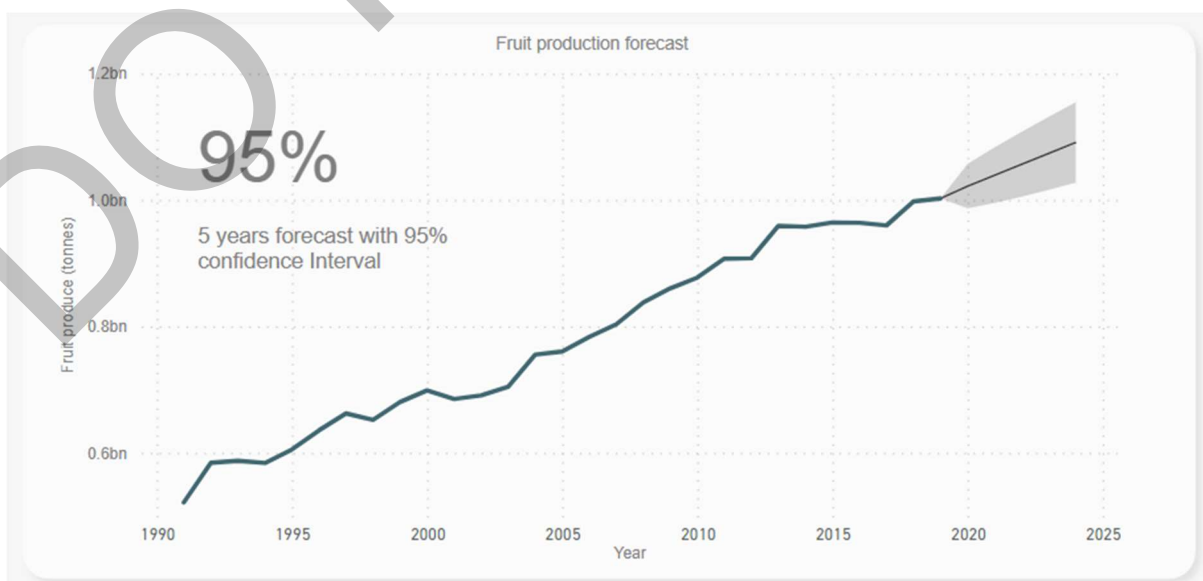
The area chart above gives a quick overview of the total fruit and vegetable production in tonnes globally between the year 1991 and 2019. The chart also changes when a particular country, group of countries or continent are selected from the filters above. Apple among other fruit types is the most produced globally, while avocado is the least produced.

- iv. What is the total and average fruit production (tonnes) and the rank of a country in terms of this produce?



The cards above respectively show the Overall (global) ranking of a selected country in terms of fruit and vegetable production, the continental ranking, and the total sum of fruit and vegetables produced between the year 1991 to 2019.

- v. What will the estimated total fruit production be in five years?



This chart shows the total fruit production throughout the year 1991 to 2019 and also the forecast of future values for this production (5 years) with a 95% confidence interval. The global fruit and vegetable production will continue to rise and will exceed 1.6 billion tonnes by 2022.

Filters used

Page Filters

Continent

All

Country

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Year

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Top Bottom

The Top and Bottom buttons above enable the switch (ctrl + click) between the top and least producers of the different fruits and vegetables, the filters give the user the leverage to select any country, continent and year or year range (hold ctrl) to analyse.

d) ROOT AND TUBER ANALYSIS

- i. Which country or countries are the top and least producers of root and tuber crops and what are the percentage changes in production?

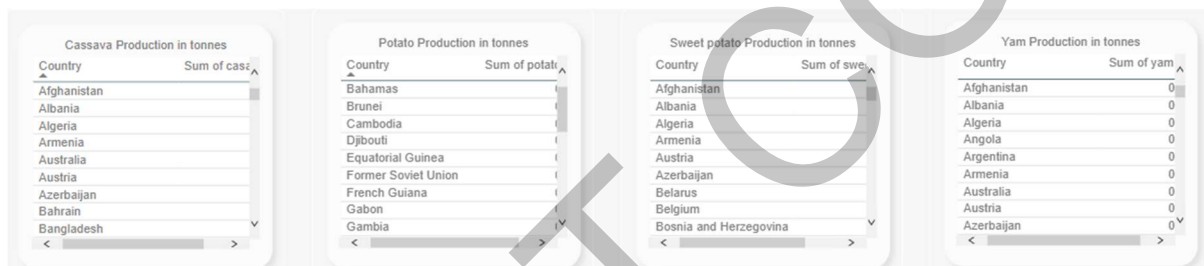
Top 5 producers of roots and tuber crops and the global percentage change from 1991-2019



This chart shows the top 5 producers of roots and tubers, and the percentage change in production. The percentage change can be selected for different year ranges, but the chart above shows the percentage change between 1991 and 2019.

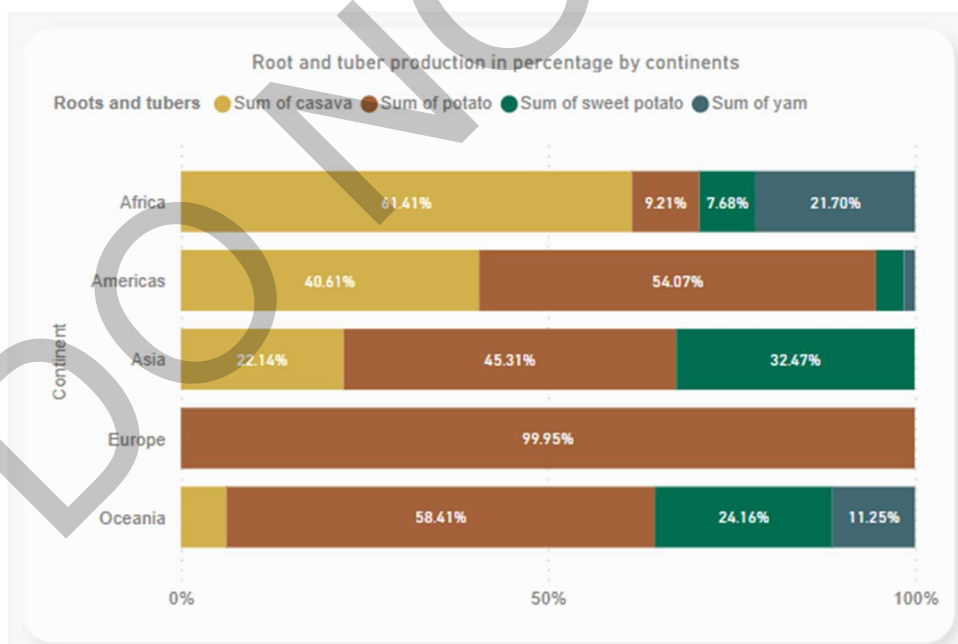
- Nigeria is the top producers of yam and cassava. Nigeria's cassava produce is almost double the total cassava production of Thailand, the second highest producers of cassava and its yam production is almost 7 time that of Ghana the second top producers of yam.
- China is the top producers of potato and sweet potato. China's sweet potato produce is almost 32 times the total sweet potato production in Nigeria the second top producers and it produces more than double, the total potato production in India.
- There has been a decline by 32% in the global sweet potato production between 1991 and 2019, while other root and tuber crops has increased substantially.

Least producers of Root and tuber crops



The chart above shows countries with no record of significant production of Root and Tuber crops.

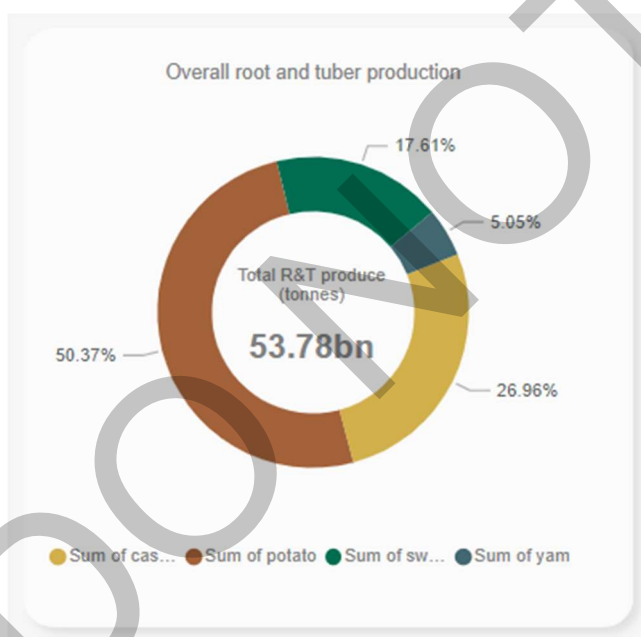
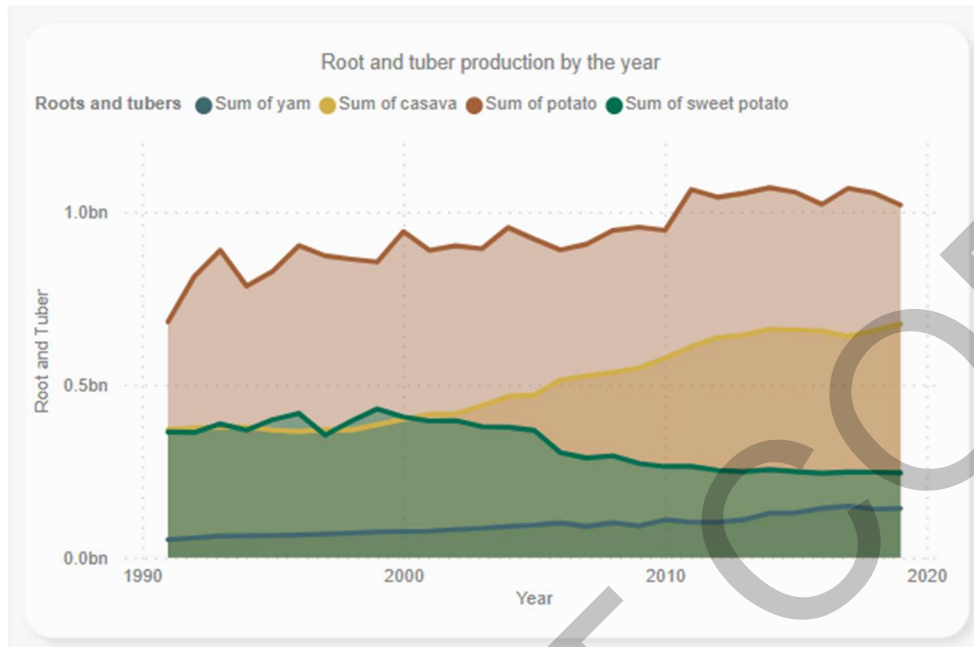
- ii. What is the distribution of root and tubers produce in percentage among continents?



This chart creates a quick summary of the distribution in percentage of Roots and tuber production by continents from 1991-2019. Cassava is the major crop produced in Africa, total potato production in Europe account for almost half of its total root and tuber. In the Americas, cassava and potato are the

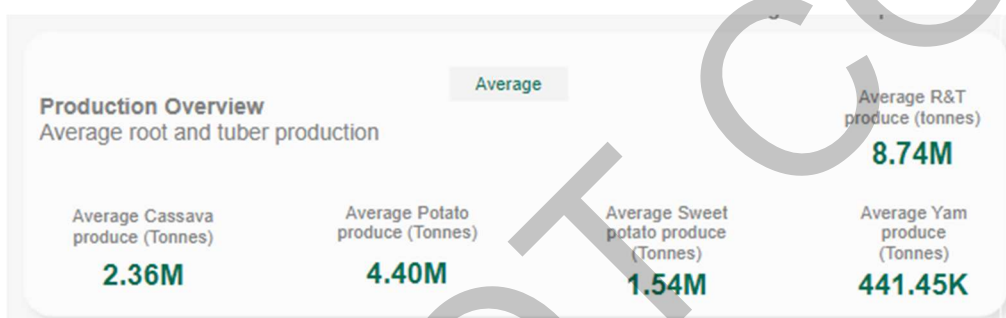
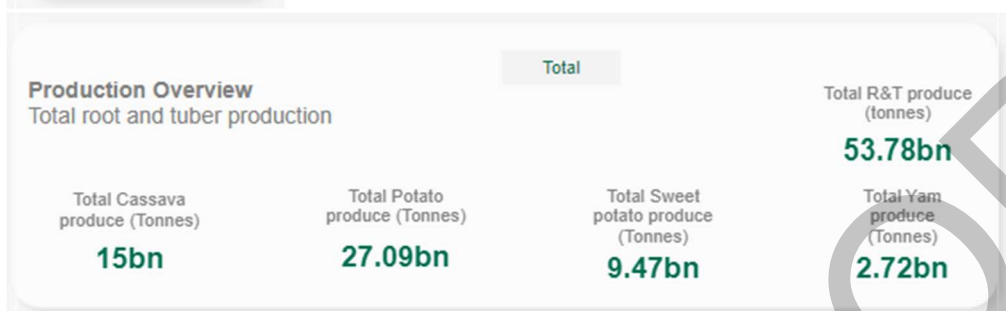
major crop produced. More than 90% of Oceania's and Europe's root and tuber produce comes from potato.

iii. Which root and vegetable type are the most produced and least produced globally?



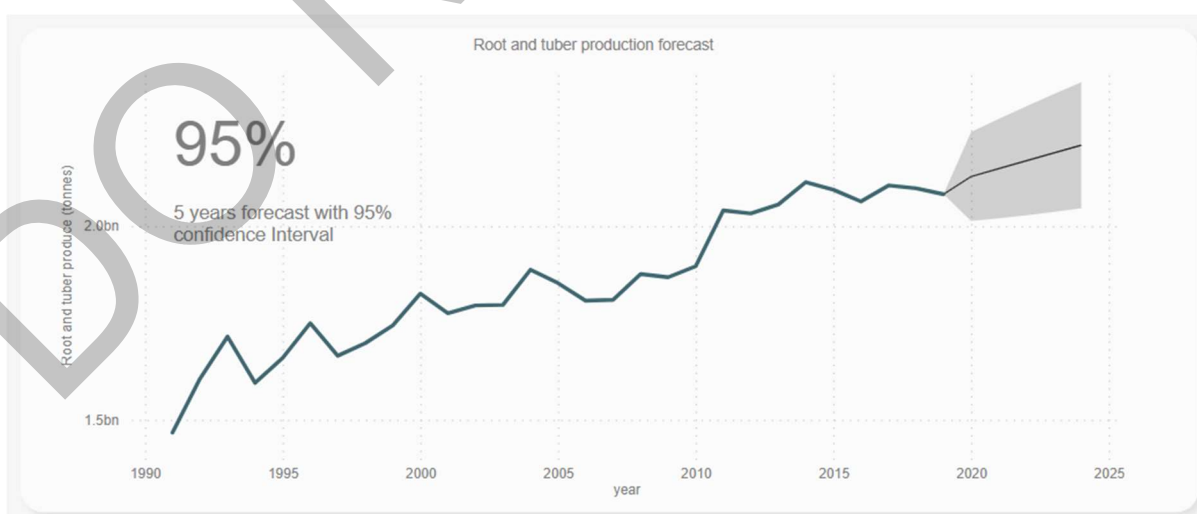
The area chart above gives a quick overview of the total root and vegetable produce in tonnes globally between the year 1991 and 2019. The chart shows that potato among other root and tuber crops is the most produced in tonnes, while yam is the least produced.

iv. What is the total and average Root and tuber production (tonnes), the rank of a country in terms of this produce and the latest Income category according to world bank outlined standard?



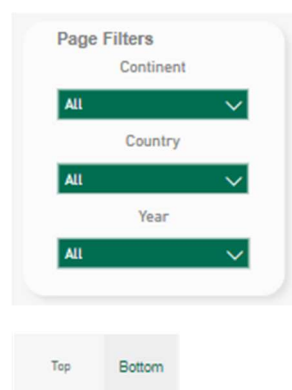
The cards above respectively show the Overall (global) ranking of a selected country in terms of root and tuber production, the continental ranking, and the total sum of roots and tubers produce between the year 1991 to 2019.

- v. What will the estimated total Root and tuber production be in five years?



This chart shows the total root and tuber produce by the year through 1991 to 2019, and the forecast of future values (5 years) with a confidence interval of 95%. The global fruit and vegetable production will continue to rise and will exceed 2.2 billion tonnes by 2024.

Filters used



Page Filters

Continent

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Country

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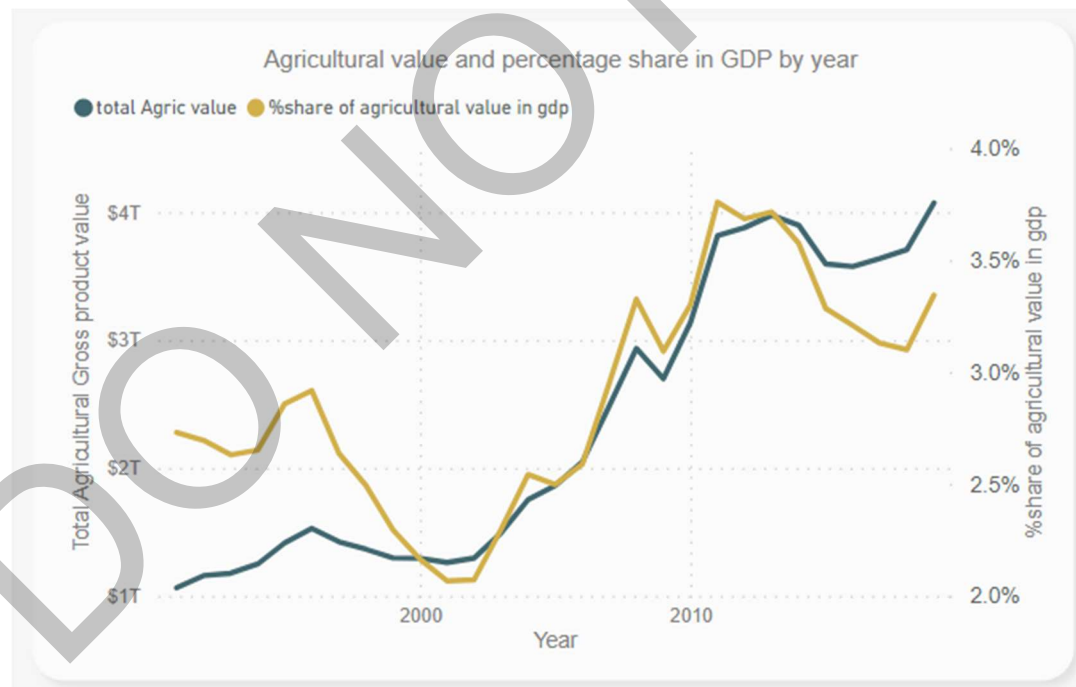
Year

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Top Bottom

The Top and Bottom buttons above enable the switch (ctrl + click) between the top and least producers of the different root and tuber crops, the filters give the user the leverage to select any country, continent and year or year range (hold ctrl) to analyse.

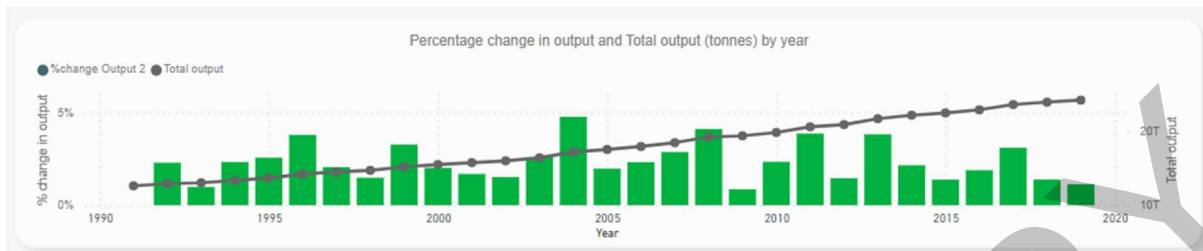
2. What are the agricultural values of a country and it's share in GDP?



The chart above shows total monetary value generated from agriculture as well as the share in GDP.

Between the year 1996 and 2002, agriculture added the least value in global gross domestic product. However, the value generated by agriculture globally and the share in GDP and shown great increase afterwards.

3. What is the percentage change in agricultural output through the year 1991 – 2019?



The line chart shows the total agricultural output in tonne globally from 1991 – 2019. The green columns indicate a positive change in percentage relative to the previous year. However, that has not been the case for Europe and Oceania. Among other continents Oceania and Europe has recorded the highest number of negative changes in percentage.

Percentage change in Europe's agricultural output by year

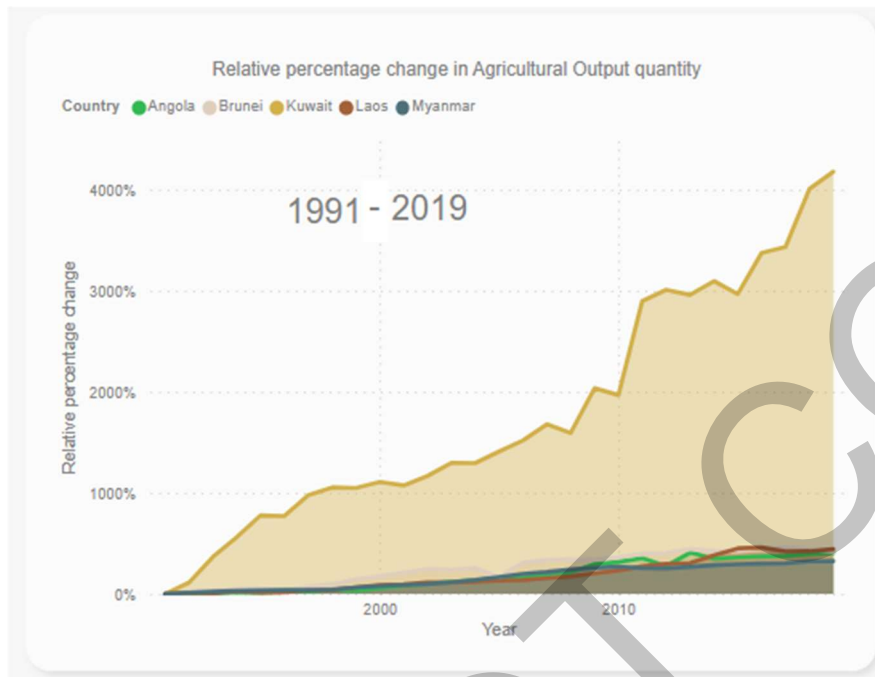


Percentage change in Oceania's agricultural output by year



4. By what percentage (larger or smaller) did the agricultural output differ from the year of reference (1991) and which country or countries has made the most significant increase?

Top 5 countries with the most significant change in agricultural output relative to the year 1991

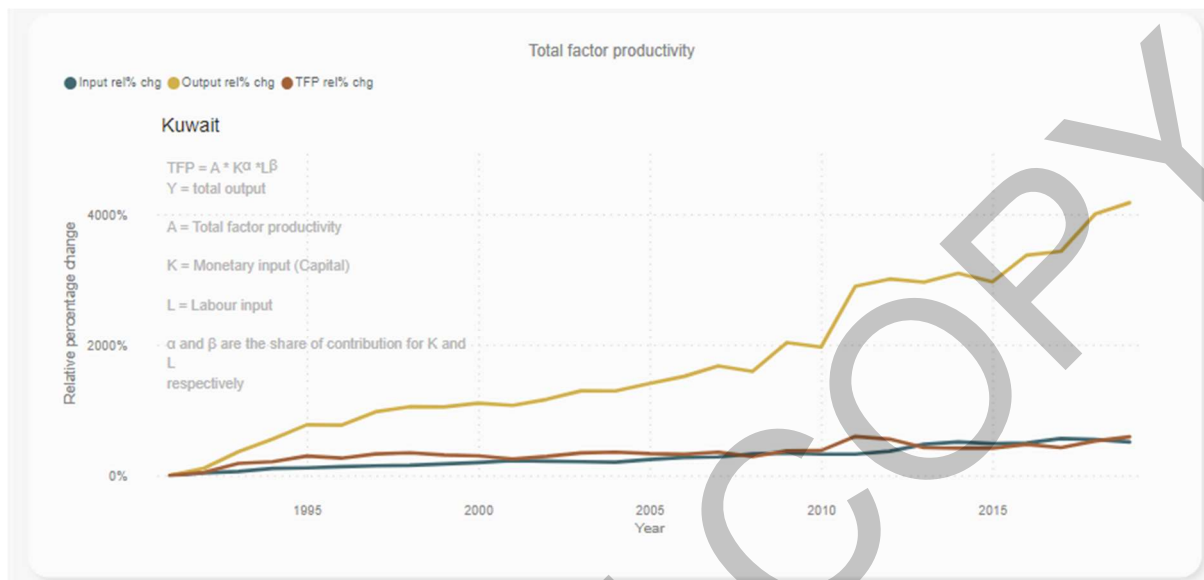


This chart gives a quick overview of countries that have managed to increase their agricultural output since the year 1991.

Kuwait among the other top 5 countries has increased its agricultural output by about 4000%, while the remaining four by more than 100%.

5. Which countries have managed to decouple agricultural output from more input?

Top 1 country to decouple agricultural output from more input.



The chart above shows the relative percentage change in input (land, labour, fertilizer, pesticides), output (crop and livestock) and the total factor productivity.

Total factor productivity is a measure of how efficiently agricultural inputs are converted into agricultural outputs. Inputs would directly match outputs if productivity did not improve. The total factor productivity goes by the formula:

$$Y = A * K^\alpha * L^\beta$$

Y = total output

A = Total factor productivity

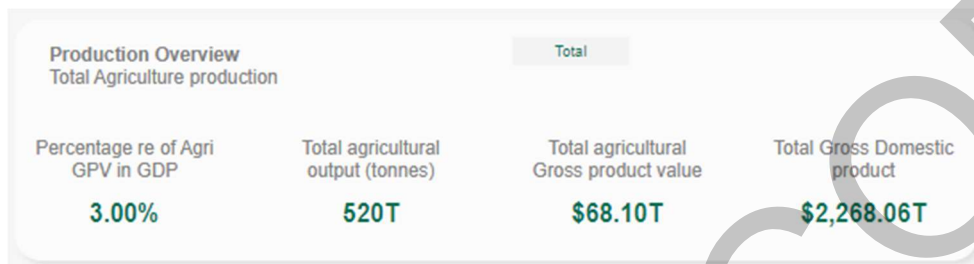
K = Monetary input (Capital)

L = Labour input

α and β are the share of contribution for K and L respectively

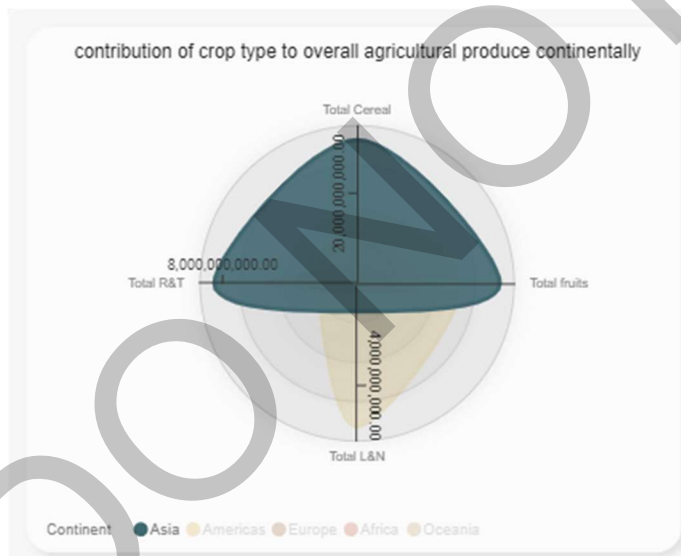
Kuwait among other countries has been able to significantly increase drastically agricultural output from less inputs.

6. What is the total and average agricultural value, agricultural output and ranking of countries in terms of agricultural output?

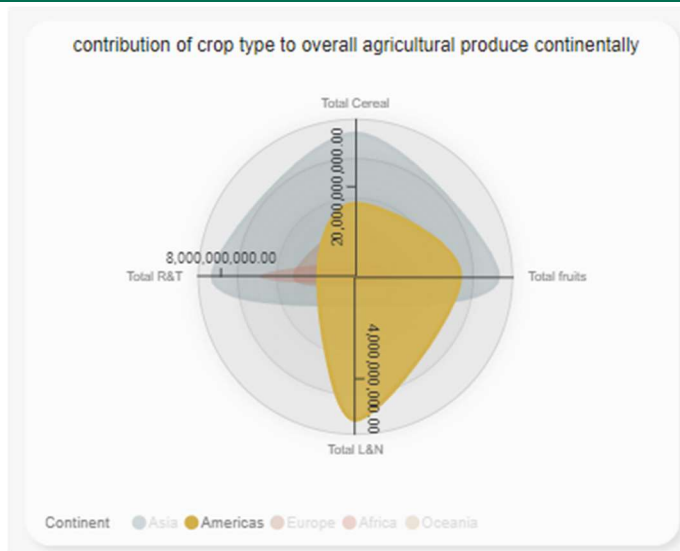


The cards above show the cumulative and average agricultural output and value globally, \$68.10 Trillion has been generated from agriculture between the year 1991 and 2019. The total output in tonnes from agriculture from 1991 to 2019 is 520 trillion.

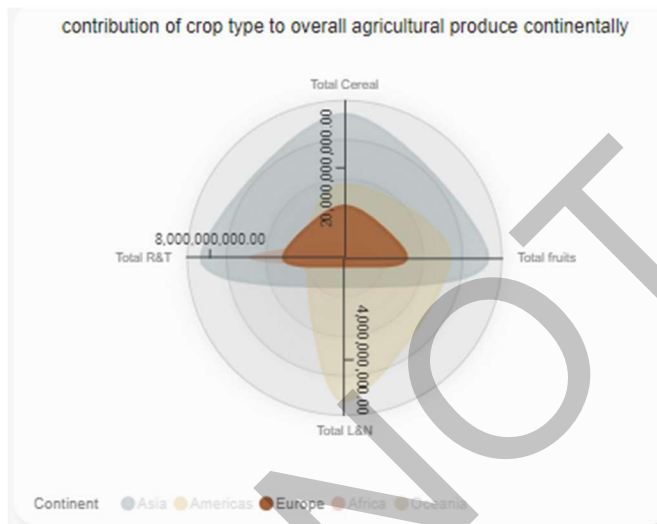
7. Which crop type is the most produced continentally?



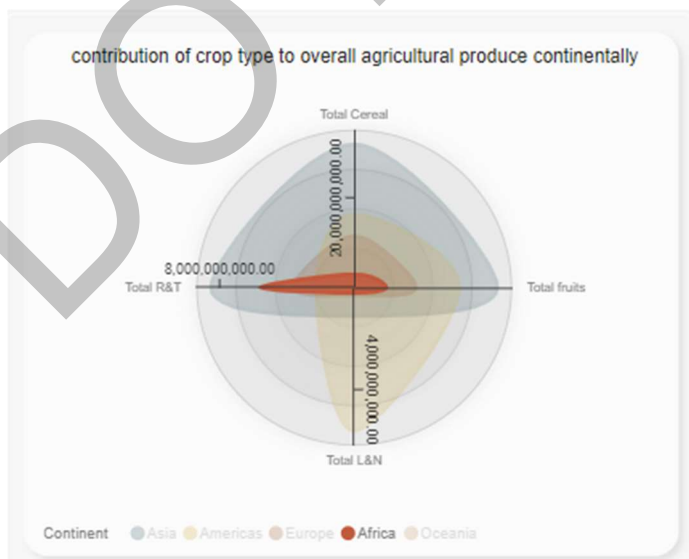
In the continent Asia which accounts for the major agricultural produce globally, Root and tubers, cereal and fruits are the most produced crop type, the production of legume and nuts is almost insignificant.



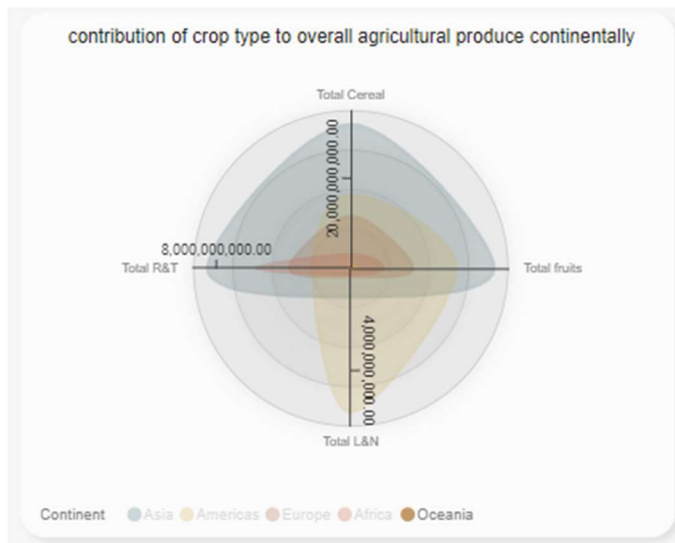
In Americas the major crop grown is Legumes and nuts, there is also a fair production in fruits and cereal.



In Europe, Cereal, fruits and root and tubers are the most produced crop type.

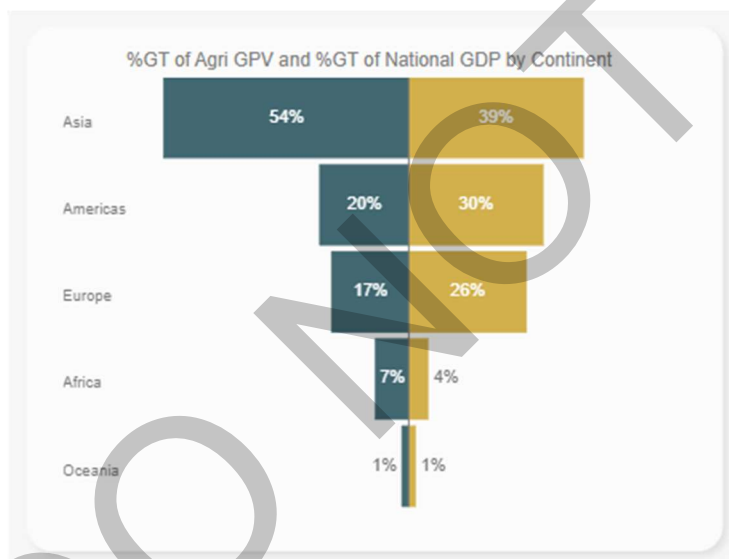


In Africa, Root and tubers lead the race in terms of agricultural produce.



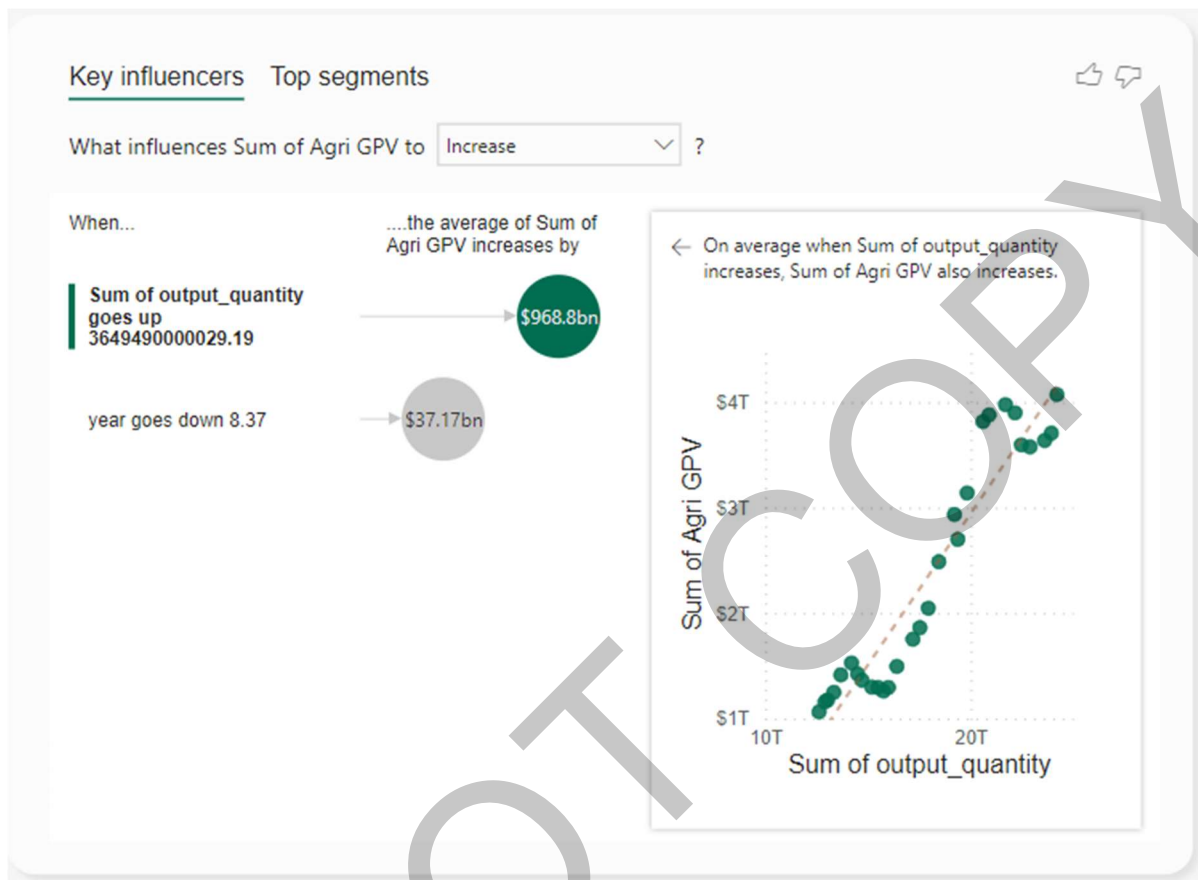
Oceania shows no sign of contribution to agricultural production.

8. What percentage of agricultural market and global economy is held by the different continents?



Asia, accounts for more than one third of global economy and over 50% of the agricultural market. America after Asia leads the race in percentage share of global economy and almost the same share in agricultural market as Europe. Oceania holds the least, one percent in both global and agricultural economy.

9. How does agricultural output and year impact agricultural gross product value?

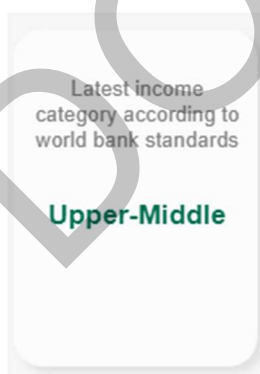


The Key influencer is one of the greatest tools created by Microsoft in power BI, it automatically interpreted with high precision in influence of a factor on another. When the sum of agricultural output globally goes up by 3.65 trillion, agricultural value increases by \$968.8 billion. Given the correlation between agricultural output quantity and value, it can be said that agricultural gross product value increases as the sum of agricultural output increases.



The year shows an inverse proportionality to Agricultural gross product value, It suggests that previous years has shown greater increase in agricultural gross product value compared to present, and further suggest that as the year increase the gross agricultural product value will like decrease.

10. What is the Latest Income category of a country as per World Bank standards?



This calculated measure automatically displays the income category of a country, upon selection.

Continent	Total Cassava	Total Potato	Total Sweet potato	
Africa	3,442,086,826.00	515,933,289.00	430,579,443.00	1,2
Americas	901,326,194.00	1,200,033,862.00	84,927,251.00	
Asia	1,891,002,717.00	3,869,731,722.00	2,773,185,741.00	
Afghanistan	0.00	9,035,861.00	0.00	
Armenia	0.00	13,884,147.00	0.00	
Azerbaijan	0.00	19,607,031.00	0.00	
Bahrain	0.00	1,362.00	177.00	
Bangladesh	0.00	148,322,691.00	9,777,840.00	
Bhutan	0.00	1,189,551.00	1,080.00	
Brunei	65,851.00	0.00	6,388.00	
Cambodia	86,502,255.00	0.00	1,231,149.00	
China	119,790,923.00	1,948,225,531.00	2,544,769,571.00	
Cyprus	0.00	3,880,626.00	0.00	
Total	6,239,663,032.00	9,263,694,475.00	3,311,004,173.00	1,2

Finally, the table used in the fruit, cereal, legumes, and roots page presents information on the top and lowest producing countries and continents, with darker green indicating higher production and lighter green indicating lower production.

Conclusion

In conclusion, the top producers of legumes and nuts are India, Canada, the United States, Brazil, and Nigeria, with India being the top producer of beans and cassava, Canada the top producer of peas, and the United States the top producer of soybeans. The production of cashews and soybeans has significantly increased since 1991. Soybeans are the most produced legume and nut globally, with more than 70% of total production coming from America and Asia, and more than 50% coming from Europe and Oceania. The top producers of cereal crops are China, Russia, the United States, India, and France, with the United States being the top producer of maize, China the top producer of wheat and rice, and Russia the top producer of rye and barley. Cereal production is distributed differently among continents, with Oceania producing mostly wheat, America producing mostly maize, and Europe, Africa, and Asia producing mostly wheat, rice, and maize. The top producers of oil crops are the United States, China, Brazil, Indonesia, and India, with the United States being the top producer of soybeans, China the top producer of rapeseed, and India the top producer of cottonseed. Soybeans are the most produced oil crop globally, with more than 50% of total production coming from America, Europe, and Asia, and more than 50% coming from Africa and Oceania.

Recommendations

1. Encouraging the expansion of soybean production in countries that are already top producers, as it is the most widely produced legume and nut globally.
2. Identifying opportunities for the expansion of production of legumes and nuts in countries that are currently not significant producers, to increase global production and potentially improve food security in those regions.
3. Supporting the expansion of cereal production in countries that are already top producers, particularly for crops that are in high demand globally, such as wheat and rice.
4. Encouraging the development of new technologies or techniques that could increase the efficiency and productivity of legume, nut, cereal, and fruit production in various countries.
5. Adoption of strategy used by countries like Kuwait in decoupling agricultural outputs.

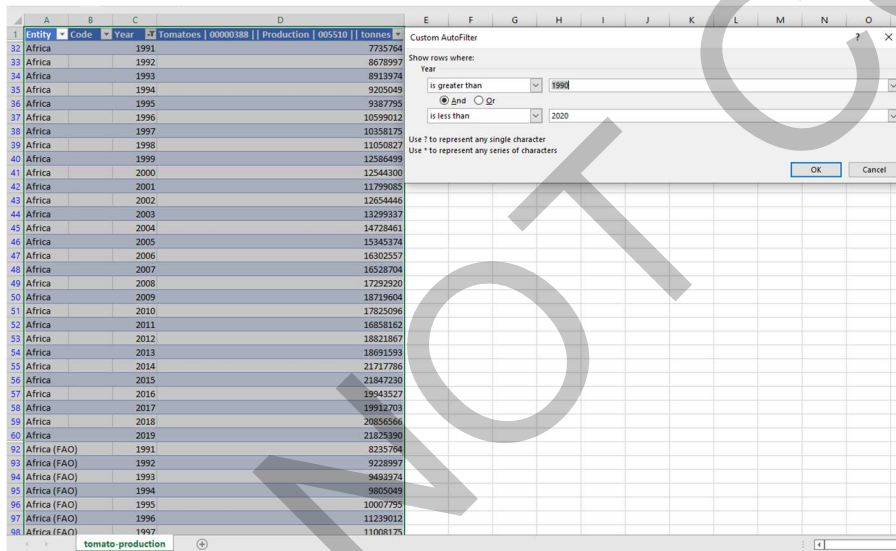
Appendix

1) Data Pre-processing

The dataset was prepared in Excel, and power BI. A little transformation was done using R. For the sake of consistency, this dataset only considers data from the year 1991 to 2019. The pre-processing steps was composed of four major processes, they include:

- Year Filter
- R scripting for continent column
- Creating the fact table (country/continent table)
- Merging of columns and Tables
- Handling null values
- Creation of date column
- Renaming columns

a) Year Filter



Entity	Code	Year	Tomatoes (000000)	Production (000000)
Africa		1991	7735764	
Africa		1992	8678997	
Africa		1993	8913974	
Africa		1994	9205049	
Africa		1995	9387795	
Africa		1996	10599012	
Africa		1997	10358175	
Africa		1998	11050827	
Africa		1999	12586499	
Africa		2000	12544300	
Africa		2001	11799085	
Africa		2002	12154446	
Africa		2003	13299337	
Africa		2004	14728461	
Africa		2005	15345374	
Africa		2006	16302557	
Africa		2007	16528704	
Africa		2008	17292920	
Africa		2009	18719804	
Africa		2010	17825096	
Africa		2011	16858182	
Africa		2012	18821867	
Africa		2013	18691593	
Africa		2014	21717786	
Africa		2015	21847230	
Africa		2016	19415527	
Africa		2017	19912703	
Africa		2018	20856566	
Africa		2019	21025390	
Africa (FAO)		1991	8235764	
Africa (FAO)		1992	9228997	
Africa (FAO)		1993	9493974	
Africa (FAO)		1994	9805049	
Africa (FAO)		1995	10007795	
Africa (FAO)		1996	11239012	
Africa (FAO)		1997	11008175	

The data was processed, and the year was filtered to only contain data from the selected year. Afterwards, the data used for the analysis was then assembled in a single Excel worksheet.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Entity	Code	Year	number employed agri												
2	Afghanistan: AFG		1991	2092348.284												
3	Afghanistan: AFG		1992	2296014.044												
4	Afghanistan: AFG		1993	2542180.4												
5	Afghanistan: AFG		1994	2750005.458												
6	Afghanistan: AFG		1995	2918585.497												
7	Afghanistan: AFG		1996	3037002.198												
8	Afghanistan: AFG		1997	3111408.26												
9	Afghanistan: AFG		1998	3164596.604												
10	Afghanistan: AFG		1999	3233335.122												
11	Afghanistan: AFG		2000	3338758.724												
12	Afghanistan: AFG		2001	3509000.875												
13	Afghanistan: AFG		2002	3590422.186												
14	Afghanistan: AFG		2003	3749959.804												
15	Afghanistan: AFG		2004	3921699.87												
16	Afghanistan: AFG		2005	4029469.868												
17	Afghanistan: AFG		2006	4071664.34												
18	Afghanistan: AFG		2007	4071698.673												
19	Afghanistan: AFG		2008	4083787.948												
20	Afghanistan: AFG		2009	3998362.28												
21	Afghanistan: AFG		2010	3937605.461												
22	Afghanistan: AFG		2011	4029127.694												
23	Afghanistan: AFG		2012	4055946.365												
24	Afghanistan: AFG		2013	4138968.851												
25	Afghanistan: AFG		2014	4231880.678												
26	Afghanistan: AFG		2015	4318997.725												
27	Afghanistan: AFG		2016	4382769.725												
28	Afghanistan: AFG		2017	4376415.701												
29	Afghanistan: AFG		2018	4450650.799												
30	Afghanistan: AFG		2019	4548158.325												
31	Africa Eastern and S		1991	82100218.52												
32	Africa Eastern and S		1992	84561738.18												
33	Africa Eastern and S		1993	86959362.62												
34	Africa Eastern and S		1994	89472344.47												
35	Africa Eastern and S		1995	92014600.89												
36	Africa Eastern and S		1996	94275577.56												
37	Africa Eastern and S		1997	96815014.9												
38	Africa Eastern and S		1998	99480072.29												

b) R scripting for continent column

Run R script

Enter R scripts into the editor to transform and shape your data.

Script

```
df <- dataset
library(countrycode)
df$Continent <- countrycode(sourcevar = df$Entity,
origin = 'country.name', destination = 'continent')
```

The script will run with the following R installation C:\Program Files\Microsoft R Open\R-4.0.2.

To configure your settings and change which R installation you want to run, go to Options and settings.

OK

Cancel

The continents of countries were missing in the dataset. To handle this case automatically, we made use of the R Script on the output quantity table. Power BI allows us the easily run script file without having to worry about the current name of the power query step which it considers the dataset, instead it allows us access the dataset with a custom name 'dataset'.

- The library "countrycode" allows the appending of Continents to countries.
- The code "df\$Continent" – creates a new column Continent
- <- - for assignment
- "sourcevar" - in the "countrycode" class, allows us to specify the country column (df\$Entity)
- "origin" and "destination" - specifies the format for conversion, in this case (country to continent)

c) Creating the most important dimension tables (country/continent and year)

After the R transformation, a duplicate of the output quantity table was created to form the country/continent and year table.

- Creating the country/continent table

The other columns were removed from the table excluding the continent and country column. The duplicates were removed from the country column and not the continent column, this is to ensure successful mapping of each country to its continent for later sub setting. An index column was created afterwards from the add column tab.

123 Country Index	A ^B _C Country	A ^B _C Continent
1	Albania	Europe
2	Algeria	Africa
3	Angola	Africa
4	Antigua and Barbuda	Americas
5	Argentina	Americas
6	Armenia	Asia
7	Australia	Oceania
8	Austria	Europe
9	Azerbaijan	Asia
10	Bahrain	Asia
11	Bangladesh	Asia
12	Barbados	Americas
13	Belarus	Europe
14	Belgium	Europe
15	Belize	Americas
16	Benin	Africa
17	Bhutan	Asia
18	Bolivia	Americas

After the creation of the country and continent dimension table, another table was then created to contain only the continents without duplicates. An index column was created afterwards from the add column tab.

i^2_3 Continent index	A^B_C Continent
1	Europe
2	Africa
3	Americas
4	Asia
5	Oceania

- Creating the year column

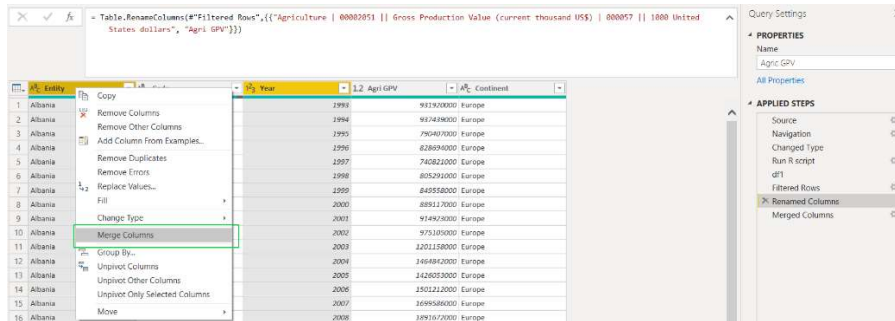
Another duplicate was created from the output quantity column and only the year column was kept. Duplicates of the year column were removed, leaving years between 1991 and 2019. An index column was created afterwards.

i^2_3 Year index	i^2_3 year
1	1991
2	1992
3	1993
4	1994
5	1995
6	1996
7	1997
8	1998
9	1999
10	2000
11	2001
12	2002
13	2003
14	2004
15	2005

d) Merging of columns and Tables

In this step, various independent columns were merged for easy analysis. The Agric GPV and National GDP, Employment and National GDP, Fruits and Vegetable tables, Root and tuber tables, Legumes and nuts table and cereal table were merged to get distinct tables that represented the various individual tables. An example of the merge process is shown below using the Agric GPV and National GDP.

The dataset contains data for different countries at different years (1991 – 2019). Therefore, to merge columns based on this fact, the only realistic method is to merge the country (Entity) and year column into one column ‘Entity year’ for the two tables before merging them based on this.



The screenshot shows the Power Query Editor interface. The main area displays a table with columns: Entity, Year, L2 Agri GPV, and L2 Continent. A context menu is open over the 'Entity' column, with 'Merge Columns' highlighted. The 'Query Settings' pane on the right shows the 'Properties' tab with the name 'Agri GPV' and a list of applied steps including 'Source', 'Navigation', 'Changed Type', 'Run R script', 'dfl', 'Filtered Rows', and 'Renamed Columns'.

```

= Table.CombineColumns(Table.TransformColumnTypes(#"Renamed Columns", {{"Year", type text}}, "en-GB"), {"Entity", "Year"},
    Combiner.CombineTextByDelimiter(",", QuoteStyle.None), "Entity year")

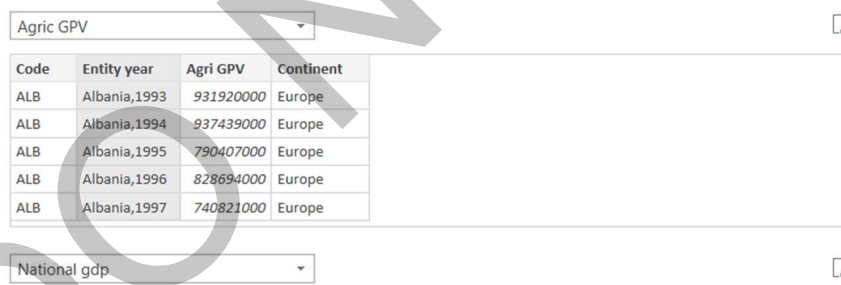
```

The resulting table after the merge is shown below:

Code	Entity year	L2 Agri GPV	L2 Continent
1	ALB Albania,1993	931920000	Europe
2	ALB Albania,1994	937439000	Europe
3	ALB Albania,1995	790407000	Europe
4	ALB Albania,1996	828694000	Europe
5	ALB Albania,1997	740821000	Europe
6	ALB Albania,1998	805291000	Europe
7	ALB Albania,1999	849558000	Europe
8	ALB Albania,2000	889117000	Europe

Afterwards, the “Merge queries” function were then used to merge the various tables and the “Entity” and “year” were then split by delimiter (“,”) into two separate columns after the process.

Select tables and matching columns to create a merged table.



The screenshot shows the 'Merge Queries' dialog in Power Query. The 'Agri GPV' table is selected from the dropdown. The 'National gdp' table is also selected. The dialog shows the columns for each table: 'Code', 'Entity year', 'Agri GPV', and 'Continent' for the first table, and 'Code', 'Entity year', 'GDP (output, multiple price benchmarks)', and 'Continent' for the second table.

Code	Entity year	Agri GPV	Continent
ALB	Albania,1993	931920000	Europe
ALB	Albania,1994	937439000	Europe
ALB	Albania,1995	790407000	Europe
ALB	Albania,1996	828694000	Europe
ALB	Albania,1997	740821000	Europe

Code	Entity year	GDP (output, multiple price benchmarks)	Continent
ALB	Albania,1991	10822847656	
ALB	Albania,1992	9767368164	
ALB	Albania,1993	11222619141	
ALB	Albania,1994	12672190430	
ALB	Albania,1995	13669630859	

- Final Result

Table.RemoveColumns("#Reordered Columns",{"Code"})					
Entity	Continent	year	Agri GVP	National.gdp.GDP (output, multiple price benchmarks)	
1 Albania	Europe	1993	931920000	11222619141	
2 Albania	Europe	1994	937439000	12672190430	
3 Albania	Europe	1995	790407000	13669630859	
4 Albania	Europe	1996	828694000	15067375000	
5 Albania	Europe	1997	740821000	14072309570	
6 Albania	Europe	1998	805291000	15531704102	
7 Albania	Europe	1999	849558000	16525398438	
8 Albania	Europe	2000	889117000	16633271484	
9 Albania	Europe	2001	914923000	17293777344	
10 Albania	Europe	2002	975105000	17619767578	
11 Albania	Europe	2003	1201158000	18275126953	
12 Albania	Europe	2004	1464842000	18750121094	
13 Albania	Europe	2005	1426053000	20162416016	
14 Albania	Europe	2006	1501212000	21773833984	
15 Albania	Europe	2007	1699586000	23260232422	
16 Albania	Europe	2008	1891672000	25868791016	
17 Albania	Europe	2009	1841674000	27709900391	
18 Albania	Europe	2010	1903675000	31021064453	

e) Handling null values

For the datasets used, null value represents an insignificant amount (nearly 0) and as such were replaced with the number zero. An example is shown below used the merged cereal table; tables such as Barkley, Rye, Maize, Rice, and Wheat had null values in some countries and year. It is assumed that no significant production in tons were made for that year.

Table.RenameColumns("#Changed Type1",{"Entity year.2", "year"}, {"Entity year.1", "Entity"})						
Barley	Rye	Cereals	Maize	Rice	Wheat	
1991	217000	2721302	420000	335000	1726000	
1992	200000	2470000	302000	300000	1650000	
1993	250000	2981774	470000	300000	1940000	
1994	230000	3144000	500000	342000	2050000	
1995	200000	3242179	530000	390000	2100000	
1996	230000	3241915	360000	340000	2300000	
1997	217000	2721302	420000	335000	1726000	
1998	200000	2470000	302000	300000	1650000	
1999	250000	2981774	470000	300000	1940000	
2000	230000	3144000	500000	342000	2050000	
2001	200000	3242179	530000	390000	2100000	
2002	230000	3241915	360000	340000	2300000	
2003	217000	2721302	420000	335000	1726000	
2004	200000	2470000	302000	300000	1650000	
2005	250000	2981774	470000	300000	1940000	
2006	230000	3144000	500000	342000	2050000	
2007	200000	3242179	530000	390000	2100000	
2008	230000	3241915	360000	340000	2300000	
2009	486000	6314000	300000	645000	5064000	
2010	437000	5997000	301000	672000	4532000	
2011	305900	4681020	300120	672000	3388000	
2012	504000	6379000	310000	500000	5050000	
2013	514000	6520329	312000	512094	5169235	
2014	521000	6748023	316000	537000	5370259	
2015	403000	5808288	316000	410000	4673040	
2016	307856	5532695	311646	356565	4555110	
2017	94995	4894365	173912	338420	4280776	

f) Creation of 'Date' column

The date column was created using the year column provided. It is assumed that the data was collected on the last day of each year, hence the date was created by adding a custom column with the last day and month '31/12/' which was then merged with the year column to form the date column. Note, the newly created date column was converted to date type.

1 ² ₃ Year index	Date	1 ² ₃ year
1	31/12/1991	1991
2	31/12/1992	1992
3	31/12/1993	1993
4	31/12/1994	1994
5	31/12/1995	1995
6	31/12/1996	1996
7	31/12/1997	1997
8	31/12/1998	1998
9	31/12/1999	1999
10	31/12/2000	2000
11	31/12/2001	2001
12	31/12/2002	2002
13	31/12/2003	2003
14	31/12/2004	2004
15	31/12/2005	2005
16	31/12/2006	2006
17	31/12/2007	2007
18	31/12/2008	2008

g) Renaming columns

Finally, to ensure uniformity and simplicity across the dataset used, some columns were renamed to maintain uniformity example the 'country' column while columns with long names were shortened.

2) Data Modelling (Snowflake schema)

As a requirement for the ICA, we have been asked to make a detailed explanation of the type of power bi model implemented. Firstly, kindly note that this case study is a novel one that does not necessarily apply to common business practices except for companies with large data warehouses dealing with more than one fact tables. However, this study considers the necessary steps that should be taken should this case arise. The fact tables contain the datasets of various agricultural topics of the different countries of the world, but they also contain missing values for most of the countries. To handle this query a first merge was performed on the various topic related dataset such as cereal (contains the various data for cereal crops), Legumes and nuts (contains the various legume and nut crops) etc. Afterwards, a dimension tables of the country, continent and year was generated to serve as a link to the various fact tables. Below are the various tables used to create the agricultural topic table.

Table Name	Dimensions
Apple (fruit and veg)	Apples, Code, Entity year
Barley (cereal)	Barley, Entity year
Cashew (Legumes an...)	Cashew nuts, Entity year
casava (root and tuber)	Cassava, Entity year
National gdp	Code, Entity year, GDP (output, multiple ...)
Avocado (fruit and veg)	Avocados, Code, Entity year
cereal (cereals)	Cereals, Entity year
Beans (legumes and ...)	Beans, Entity year
potato (root and tuber)	Entity year, Potatoes
number-of-people-e...	Code, Entity year, number employed agri
Grapes (fruit and veg)	Code, Entity year, Grapes
maize (cereal)	Entity year, Maize
peas (Legumes and n...)	Entity year, Peas
sweet potato (root an...)	Entity year, Sweet potatoes
Agric GPV	Agri GPV, Code, Continent
Bananas (Fruits and v...)	Bananas, Code, Entity year
rice (cereal)	Entity year, Rice
Soybeans (legumes ...)	Entity year, Soybeans
yam (root and tuber)	Entity year, Yams
GNI per capita	Entity year, Value
Oranges (fruit and veg)	Code, Entity year, Oranges 00000490 ...
Rye (cereal)	Entity year, Rye
wheat (cereal)	Entity year, Wheat

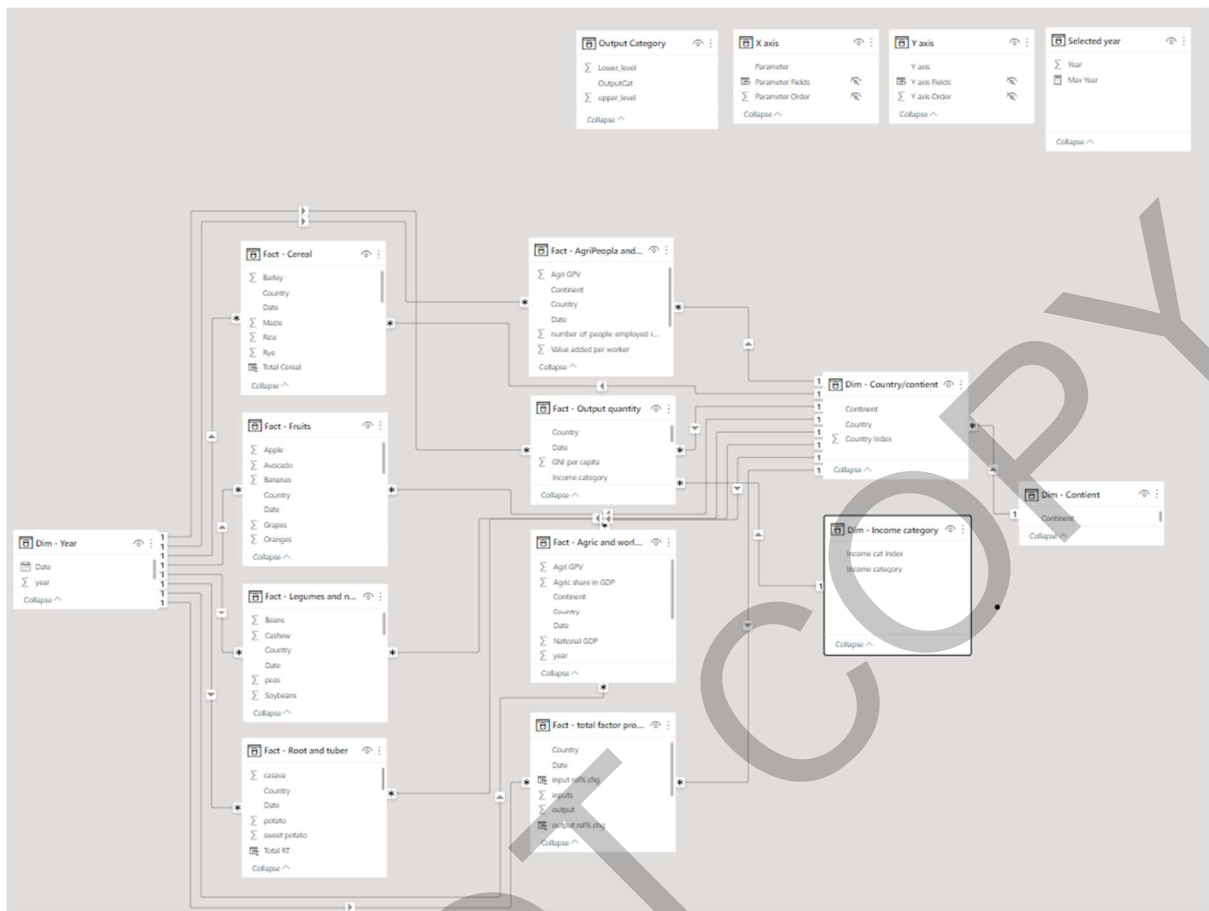
Below is the categorized tables (fact) and dimension tables. The fact tables listed were created by merging the individual tables shown above to form a group. The highlighted tables are the field parameter tables. It is discussed later in this report.

Note: the merge was performed on individual data tables such as (Rye, wheat, barley etc to create the cereal table, grape, avocado, apple etc to create the fruit table and so on) this was done to prevent major data loss when merging the tables rather than merging them all together.



Finally, the table below shows the link between the various fact tables and the dimension table. The Country and year dimension tables were created. The continental dimension table of the country/continent table was also created to hold the unique continents.

- The country/continent dimension table links all the fact tables through the country column
- The year table, links all the fact table through the Date column
- The continent table links the country/continent table through the continent column
- The Dim - Income category links the output quantity fact table and holds the unique values of the different Income groups.



This is a snowflake data model because of the sub dimension table of the country/continent table.

3) Dax and M language

Dax

In the creation of the various measures and calculated columns using DAX, rather than organising the measures in a single table, I chose to leave them in their individual source tables. This was to avoid unnecessary confusion during their application or identification. Below are the list of measures and calculated columns created for this report.

a) 'Total' calculated columns

```
total_fruits = 'Fact - Fruits and Vegetables'[Apple] + 'Fact - Fruits and Vegetables'[Avocado] + 'Fact - Fruits and Vegetables'[Bananas] + 'Fact - Fruits and Vegetables'[Grapes] + 'Fact - Fruits and Vegetables'[Oranges]
```

```
Total RT = 'Fact - Root and tuber'[casava] + 'Fact - Root and tuber'[potato] + 'Fact - Root and tuber'[sweet potato] + 'Fact - Root and tuber'[yam]
```

```
Total LN = 'Fact - Legumes and nuts'[Beans] + 'Fact - Legumes and nuts'[Cashew] + 'Fact - Legumes and nuts'[Soybeans] + 'Fact - Legumes and nuts'[peas]
```

```
Total Cereal = 'Fact - Cereal'[Barley] + 'Fact - Cereal'[Rye] + 'Fact - Cereal'[Maize] + 'Fact - Cereal'[Rice] + 'Fact - Cereal'[Wheat]
```

The total of the different crop types was calculated by summing up the individual contributing columns with the '+' sign.

b) 'Total' calculated measures

The total measures are calculated using the SUM function. Once a numeric column is passed to a SUM function, it automatically aggregates the total of the values of that column.

```
total Agric value = SUM('Fact - Agric and world GDP'[Agri GPV])

Total National GDP = SUM('Fact - Agric and world GDP'[National GDP ])

Total C = SUM('Fact - Cereal'[Total Cereal])

Total fruits = SUM('Fact - Fruits and Vegetables'[total_fruits])

Total L&N = SUM('Fact - Legumes and nuts'[Total LN])

Total output =
SUM('Fact - Output quantity'[output_quantity])

Total R&T = SUM('Fact - Root and tuber'[Total RT])

Total Barley = SUM('Fact - Cereal'[Barley])

Total Maize = SUM('Fact - Cereal'[Maize])

Total Rice = SUM('Fact - Cereal'[Rice])

Total Rye = SUM('Fact - Cereal'[Rye])

Total Wheat = SUM('Fact - Cereal'[Wheat])

Total Apple = SUM('Fact - Fruits and Vegetables'[Apple])

Total Avocado = SUM('Fact - Fruits and Vegetables'[Avocado])

Total Banana = SUM('Fact - Fruits and Vegetables'[Bananas])

Total Grape = SUM('Fact - Fruits and Vegetables'[Grapes])

Total Orange = SUM('Fact - Fruits and Vegetables'[Oranges])

Total Beans = SUM('Fact - Legumes and nuts'[Beans])

Total Cashew = SUM('Fact - Legumes and nuts'[Cashew])

Total Peas = SUM('Fact - Legumes and nuts'[peas])

Total Soybeans = SUM('Fact - Legumes and nuts'[Soybeans])

Total Cassava = SUM('Fact - Root and tuber'[casava])

Total Potato = SUM('Fact - Root and tuber'[potato])

Total Sweet potato = SUM('Fact - Root and tuber'[sweet potato])

Total yam = SUM('Fact - Root and tuber'[yam])

Total Employment = SUM('Fact - AgriPeopla and GDP'[number-of-people-employed-in-agri])
```

c) Average calculated measure

The average measures are calculated using the AVERAGE function. Once a numeric column is passed to a AVERAGE function, it automatically aggregates the AVERAGE of the values of that column using the formula below.

SUM (column values)/Number of data points.

```
Average Barley = AVERAGE('Fact - Cereal'[Barley])

Average Maize = AVERAGE('Fact - Cereal'[Maize])

Average Rice = AVERAGE('Fact - Cereal'[Rice])

Average Rye = AVERAGE('Fact - Cereal'[Rye])

Average Wheat = AVERAGE('Fact - Cereal'[Wheat])

Average Apple = AVERAGE('Fact - Fruits and Vegetables'[Apple])

Average Avocado = AVERAGE('Fact - Fruits and Vegetables'[Avocado])

Average Banana = AVERAGE('Fact - Fruits and Vegetables'[Bananas])

Average Grape = AVERAGE('Fact - Fruits and Vegetables'[Grapes])

Average Orange = AVERAGE('Fact - Fruits and Vegetables'[Oranges])

Average Beans = AVERAGE('Fact - Legumes and nuts'[Beans])

Average Cashew = AVERAGE('Fact - Legumes and nuts'[Cashew])

Average Peas = AVERAGE('Fact - Legumes and nuts'[peas])

Average Soybeans = AVERAGE('Fact - Legumes and nuts'[Soybeans])

Average Cassava = AVERAGE('Fact - Root and tuber'[casava])

Average Potato = AVERAGE('Fact - Root and tuber'[potato])

Average Sweet potato = AVERAGE('Fact - Root and tuber'[sweet potato])

Average yam = AVERAGE('Fact - Root and tuber'[yam])

Average R&T = AVERAGE('Fact - Root and tuber'[Total RT])

Average L&N = AVERAGE('Fact - Legumes and nuts'[Total LN])

Average fruits = AVERAGE('Fact - Fruits'[total_fruits])

Average C = AVERAGE('Fact - Cereal'[Total Cereal])

Average Agric GPV = AVERAGE('Fact - Agric and world GDP'[Agri GPV])

Average Employment = AVERAGE('Fact - AgriPeopla and GDP'[number-of-people-employed-in-agri])

Average output =
AVERAGE('Fact - Output quantity'[output_quantity])
```

d) Output category

The GNI per capita column was used in classifying countries into income category by looking up where the GNI per capita value lies in the Output category table. The variable Category variable hold the GNI per capita values and the CALCULATE function is then used to classify a country into an income category when the value lies between a specified Lower level and upper level.

```
1 Output Category =  
2 VAR Categoryvariable = [GNI per capita]  
3 RETURN  
4 CALCULATE(VALUE('Output Category'[OutputCat]), Categoryvariable > 'Output Category'[Lower_level], Categoryvariable <= 'Output Category'[upper_level])
```

e) Percentage change (Card)

The percentage change measure for beans, cashew, peas, soybeans, cassava, potato, sweet potato, and yam was calculated by using two CALCULATE function wrapped in a DIVIDE function. The first CALCULATE function filters the maximum selected year from the year slicer and sums up the value of that year for all countries, while the second filters the minimum and sums it up. The divide function then divides the value of the first CALCULATE function by the value of the second to produce the percentage change between selected years. If no year is selected from the year filter, it automatically calculates the percentage change between the year 1991(minimum) and 2019(maximum).

```
1 % Change Beans =  
2 DIVIDE(  
3 CALCULATE(  
4 SUM('Fact - Legumes and nuts'[Beans]),  
5 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MAX('Fact - Legumes and nuts'[Date]))  
6 )  
7 ),  
8 CALCULATE(  
9 SUM('Fact - Legumes and nuts'[Beans]),  
10 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MIN('Fact - Legumes and nuts'[Date])))  
11 ,0)  
12 - 1  
  
1 % Change Cashew =  
2 DIVIDE(  
3 CALCULATE(  
4 SUM('Fact - Legumes and nuts'[Cashew]),  
5 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MAX('Fact - Legumes and nuts'[Date]))  
6 )  
7 ),  
8 CALCULATE(  
9 SUM('Fact - Legumes and nuts'[Cashew]),  
10 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MIN('Fact - Legumes and nuts'[Date])))  
11 ,0)  
12 - 1  
  
1 % Change Peas =  
2 DIVIDE(  
3 CALCULATE(  
4 SUM('Fact - Legumes and nuts'[peas]),  
5 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MAX('Fact - Legumes and nuts'[Date]))  
6 )  
7 ),  
8 CALCULATE(  
9 SUM('Fact - Legumes and nuts'[peas]),  
10 FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MIN('Fact - Legumes and nuts'[Date])))  
11 ,0)  
12 - 1
```

```

1 % Change Soybeans =
2 DIVIDE(
3   CALCULATE(
4     SUM('Fact - Legumes and nuts'[Soybeans]),
5     FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MAX('Fact - Legumes and nuts'[Date])
6   )
7 ),
8   CALCULATE(
9     SUM('Fact - Legumes and nuts'[Soybeans]),
10    FILTER('Fact - Legumes and nuts','Fact - Legumes and nuts'[Date]=MIN('Fact - Legumes and nuts'[Date])))
11 ,0)
12 - 1

```

```

1 % Change casava =
2 DIVIDE(
3   CALCULATE(
4     SUM('Fact - Root and tuber'[casava]),
5     FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MAX('Fact - Root and tuber'[year])
6   )
7 ),
8   CALCULATE(
9     SUM('Fact - Root and tuber'[casava]),
10    FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MIN('Fact - Root and tuber'[year])))
11 ,0)
12 - 1

```

```

1 % Change potato =
2 DIVIDE(
3   CALCULATE(
4     SUM('Fact - Root and tuber'[potato]),
5     FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MAX('Fact - Root and tuber'[year])
6   )
7 ),
8   CALCULATE(
9     SUM('Fact - Root and tuber'[potato]),
10    FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MIN('Fact - Root and tuber'[year])))
11 ,0)
12 - 1

```

```

1 % Change sweet potato =
2 DIVIDE(
3   CALCULATE(
4     SUM('Fact - Root and tuber'[sweet potato]),
5     FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MAX('Fact - Root and tuber'[year])
6   )
7 ),
8   CALCULATE(
9     SUM('Fact - Root and tuber'[sweet potato]),
10    FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MIN('Fact - Root and tuber'[year])))
11 ,0)
12 - 1

```

```

1 % Change yam =
2 DIVIDE(
3   CALCULATE(
4     SUM('Fact - Root and tuber'[yam]),
5     FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MAX('Fact - Root and tuber'[year])
6   )
7 ),
8   CALCULATE(
9     SUM('Fact - Root and tuber'[yam]),
10    FILTER('Fact - Root and tuber','Fact - Root and tuber'[year]=MIN('Fact - Root and tuber'[year])))
11 ,0)
12 - 1

```

f) Percentage change (chart)

This percentage change differs from the previous as it is useful in generating charts and as such does not work with plain year value (1991), instead the year value must be converted first to date format before usage.

The percentage change output measure uses the total output measure in calculating the percentage change in output relative to the previous year. The major difference between this measure and the previous is that the previous cannot be used in generating charts as it is programmed to produce just a single value, while in this case, the measure is programmed to produce the percentage change for the individual date available.

The variable (VAR) current value holds the value of the total output. The SWITCH function is used to check the scope of the visual generated, the ISINSCOPE performs the check and returns the previous value (-1) of the current date using the DATEADD function. The value is then saved as variable (VAR) previous value which is then used to calculate the percentage change by subtracting the previous value from the current value and dividing the resulting answer by the previous value using the DIVIDE function.

```

1 %change Output 2 =
2 VAR Currentvalue = [Total output]
3 VAR Previousvalue =
4 SWITCH(
5     TRUE(),
6     ISINSCOPE('Dim - Year'[Date].[Day]), CALCULATE([Total output], DATEADD('Dim - Year'[Date], -1, DAY)),
7     ISINSCOPE('Dim - Year'[Date].[Month]), CALCULATE([Total output], DATEADD('Dim - Year'[Date], -1, MONTH)),
8     ISINSCOPE('Dim - Year'[Date].[Quarter]), CALCULATE([Total output], DATEADD('Dim - Year'[Date], -1, QUARTER)),
9     ISINSCOPE('Dim - Year'[Date].[Year]), CALCULATE([Total output], DATEADD('Dim - Year'[Date], -1, YEAR))
10 )
11 )
12
13 RETURN
14 DIVIDE(
15     Currentvalue - Previousvalue, Previousvalue
16 )

```

g) Country world Ranking

The country world ranking measure is used to generate the rank of country globally.

```

1 country world Ranking Agric value =
2
3 RANKX(
4     ALL('Dim - Country/contient'[Country]),
5     [total Agric value],
6     [total Agric value],
7     DESC,
8     Skip
9 )

1 country world Ranking C =
2
3 RANKX(
4     ALL('Dim - Country/contient'[Country]),
5     [Total C],
6     [Total C],
7     DESC,
8     Skip
9 )

```

```

1 country world Ranking F&V =
2
3 RANKX(
4     ALL('Dim - Country/continent'[Country]),
5     [Total fruits],
6     [Total fruits],
7     DESC,
8     Skip
9 )

1 country world Ranking L&N =
2
3 RANKX(
4     ALL('Dim - Country/continent'[Country]),
5     [Total L&N],
6     [Total L&N],
7     DESC,
8     Skip
9 )

1 country world Ranking in Output =
2
3 RANKX(
4     ALL('Dim - Country/continent'[Country]),
5     [Total output],
6     [Total output],
7     DESC,
8     Skip
9 )

1 country world Ranking R&T =
2
3 RANKX(
4     ALL('Dim - Country/continent'[Country]),
5     [Total R&T],
6     [Total R&T],
7     DESC,
8     Skip
9 )

```

To calculate the global ranking of a country, the world rank measure was created. It uses the country/continent dimension table that has the unique values of all country, the RANKX function and the measure by which country is to be ranked. More information on rank function is explained [here](#).

h) Country continent ranking

The continent ranking measure is used to generate the rank of country in their various continents.

This measure differs slightly from the previous measure as it filters only the countries in the continent of the selected country before performing the rank function. The RANKX function requires a table as the first parameter, to handle this a variable Country In Continent was created to hold the table generated from the filter process using the CALCULATETABLE function. The CALCULATETABLE function returns a table after calculation. The values of the country columns were passed as a parameter to this function that depends on the filter function to return the countries of the selected continent from the selected country.

```

1 Country continent Ranking C =
2 VAR CountriesInContinent =
3   CALCULATETABLE(
4     VALUES('Dim - Country/contient'[Country]),
5     FILTER(
6       ALL('Dim - Country/contient'),
7       'Dim - Country/contient'[Continent] = SELECTEDVALUE('Dim - Country/contient'[Continent])
8     )
9   )
10 RETURN
11
12 RANKX(
13   CountriesInContinent,
14   [Total C],
15   [Total C],
16   DESC,
17   Skip
18 )

1 Country continent Ranking F&V =
2 VAR CountriesInContinent =
3   CALCULATETABLE(
4     VALUES('Dim - Country/contient'[Country]),
5     FILTER(
6       ALL('Dim - Country/contient'),
7       'Dim - Country/contient'[Continent] = SELECTEDVALUE('Dim - Country/contient'[Continent])
8     )
9   )
10 RETURN
11
12 RANKX(
13   CountriesInContinent,
14   [Total fruits],
15   [Total fruits],
16   DESC,
17   Skip
18 )

1 Country continent Ranking L&N =
2 VAR CountriesInContinent =
3   CALCULATETABLE(
4     VALUES('Dim - Country/contient'[Country]),
5     FILTER(
6       ALL('Dim - Country/contient'),
7       'Dim - Country/contient'[Continent] = SELECTEDVALUE('Dim - Country/contient'[Continent])
8     )
9   )
10 RETURN
11
12 RANKX(
13   CountriesInContinent,
14   [Total L&N],
15   [Total L&N],
16   DESC,
17   Skip
18 )

1 Country continent Ranking R&T =
2 VAR CountriesInContinent =
3   CALCULATETABLE(
4     VALUES('Dim - Country/contient'[Country]),
5     FILTER(
6       ALL('Dim - Country/contient'),
7       'Dim - Country/contient'[Continent] = SELECTEDVALUE('Dim - Country/contient'[Continent])
8     )
9   )
10 RETURN
11
12 RANKX(
13   CountriesInContinent,
14   [Total R&T],
15   [Total R&T],
16   DESC,
17   Skip
18 )

```

i) Relative percentage change

```
rel% change output = DIVIDE('Fact - Output quantity'[output_quantity],CALCULATE(SUM('Fact - Output quantity'[output_quantity]), FILTER('Fact - Output quantity', EARLIER('Fact - Output quantity'[Country]) = 'Fact - Output quantity'[Country] && 'Fact - Output quantity'[Date] = MIN('Fact - Output quantity'[Date])))) - 1
```

```
input rel% chg = DIVIDE('Fact - total factor productivity'[inputs],CALCULATE(SUM('Fact - total factor productivity'[inputs]), FILTER('Fact - total factor productivity', EARLIER('Fact - total factor productivity'[Country]) = 'Fact - total factor productivity'[Country] && 'Fact - total factor productivity'[Date] = MIN('Fact - total factor productivity'[Date])))) - 1
```

```
output rel% chg = DIVIDE('Fact - total factor productivity'[output],CALCULATE(SUM('Fact - total factor productivity'[output]), FILTER('Fact - total factor productivity', EARLIER('Fact - total factor productivity'[Country]) = 'Fact - total factor productivity'[Country] && 'Fact - total factor productivity'[Date] = MIN('Fact - total factor productivity'[Date])))) - 1
```

```
tfp rel% chg = DIVIDE('Fact - total factor productivity'[tfp],CALCULATE(SUM('Fact - total factor productivity'[tfp]), FILTER('Fact - total factor productivity', EARLIER('Fact - total factor productivity'[Country]) = 'Fact - total factor productivity'[Country] && 'Fact - total factor productivity'[Date] = MIN('Fact - total factor productivity'[Date])))) - 1
```

The relative percentage measure is used to calculate the percentage change in relation to a specific date (1991) in this case because it is the minimum date record available.

The EARLIER function helps in calculating the relative percentage change of each country individually rather than calculating blindly. The EARLIER function evaluates a column in the outer row context in case there are multiple row contexts open in the same expression.

It is like the percentage change formular. First the filter function uses the EARLIER function to get the current country and it's least(minimum) least record date available and checks if there are any other context in the entire outer row where it has been repeated. This step is performed for all possible combination and the resulting value is divided by the value of the minimum record date for that country.

M Language

M Language or data Mash-up language has been very useful in this analysis as it provides an easy way to create tables and enables one to make changes to the table easily. To categorize countries into Income category, M language has been very useful.

I. Table creation

```
= Table.FromRecords({
    [OutputCat = "Low", Lower_level = 0, upper_level = 1046],
    [OutputCat = "Lower-Middle", Lower_level = 1046, upper_level = 4095],
    [OutputCat = "Upper Middle", Lower_level = 4095, upper_level = 12695],
    [OutputCat = "High", Lower_level = 12695, upper_level = 1000000000000000]})
```

	OutputCat	Lower_level	upper_level
1	Low	0	1046
2	Lower-Middle	1046	4095
3	Upper Middle	4095	12695
4	High	12695	1.0E+15

Firstly, the TableFromRecords function is used to create table from rows of data contained in a list format. In the example above the table was created to help categorize countries using the current classification standard set up by World Bank. Note that the names of the columns must be consistent throughout the record entry.

II. Value added per worker

Custom Column

Add a column that is computed from the other columns.

New column name
Value added per worker

Custom column formula ⓘ
= [Agri GPV] /
[#"number-of-people-employed-in-ag.number_employed_agri"]

Available columns
Code
Entity
year
Agri GPV
Continent
number-of-people-employed-in-...

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

M language has been useful in creating columns such as the value added in agriculture by worker and in the creation of the Date column. The value added per worker column was derived by dividing the total agricultural value or revenue of a country by the total number of staffs employed into agriculture. The Agri GPV column is divided by the 'number of people employed in agriculture' column.

M code for the above

```
= Table.AddColumn(#"Removed Errors", "Value added per worker", each [Agri GPV] /  
[#"number-of-people-employed-in-ag.number_employed_agri"])
```

This M language code is adding a new column to a table called "Removed Errors" called "Value added per worker." The new column calculates the value added per worker by dividing the value of agricultural gross value added (Agri GPV) by the number of people employed in agriculture (number_employed_agri).

III. Creating the Date column

Custom Column

Add a column that is computed from the other columns.

New column name
Custom

Custom column formula ⓘ
= "31/12/"

Available columns
Country
Continent
year
Agri GPV
National gdp.GDP (output, multi...
Agric share in GDP
year - Copy

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

In creating the date column, a string value '31/12/' was generated by using the power query editor. The resultant value was then merged with the year column to create the date column. The year column could perform normal calculations but advanced calculation like the percentage change (chart) can only be done using a date data type.

M code for the above

```
= Table.CombineColumns("#Duplicated Column",{ "Custom", "year"},Combiner.CombineTextByDelimiter("", QuoteStyle.None) ^
, "Date")
```

This M language code combines two columns, "Custom" and "year", into a new column called "Date". The Combiner function combines the text in the two columns using the CombineTextByDelimiter function, which combines the text with no delimiter between them (specified by the "" parameter). The QuoteStyle parameter is set to "None" which means that no quotes will be added around the combined text. The resulting "Date" column is created by the Table.CombineColumns function, which takes the input table "#Duplicated Column" and the new column name "Date" as parameters.

IV. Income Category

Custom Column

Add a column that is computed from the other columns.

New column name
Custom

Custom column formula ⓘ
= if([GNI per capita] >= 0 and [GNI per capita] <= 1046) then "Low" else if([GNI per capita] >1046 and [GNI per capita] <= 4095) then "Lower-Middle" else if([GNI per capita] >4095 and [GNI per capita] <= 12695) then "Upper-Middle" else if([GNI per capita] >12695) then "High" else ""

Available columns
Country
Date
output_quantity
GNI per capita
year

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

In creating the Income category, The World Bank category benchmark was used to set the if statements in the power query to help categorize countries into their various groups. The GNI per capita column was used for this process.

M code for the above

```
= Table.AddColumn("#Renamed Columns2", "Income category", each if([GNI per capita] >= 0 and [GNI per capita] <= 1046) then "Low" else if([GNI per capita] >1046 and [GNI per capita] <= 4095) then "Lower-Middle" else if([GNI per capita] >4095 and [GNI per capita] <= 12695) then "Upper-Middle" else if([GNI per capita] >12695) then "High" else "")
```

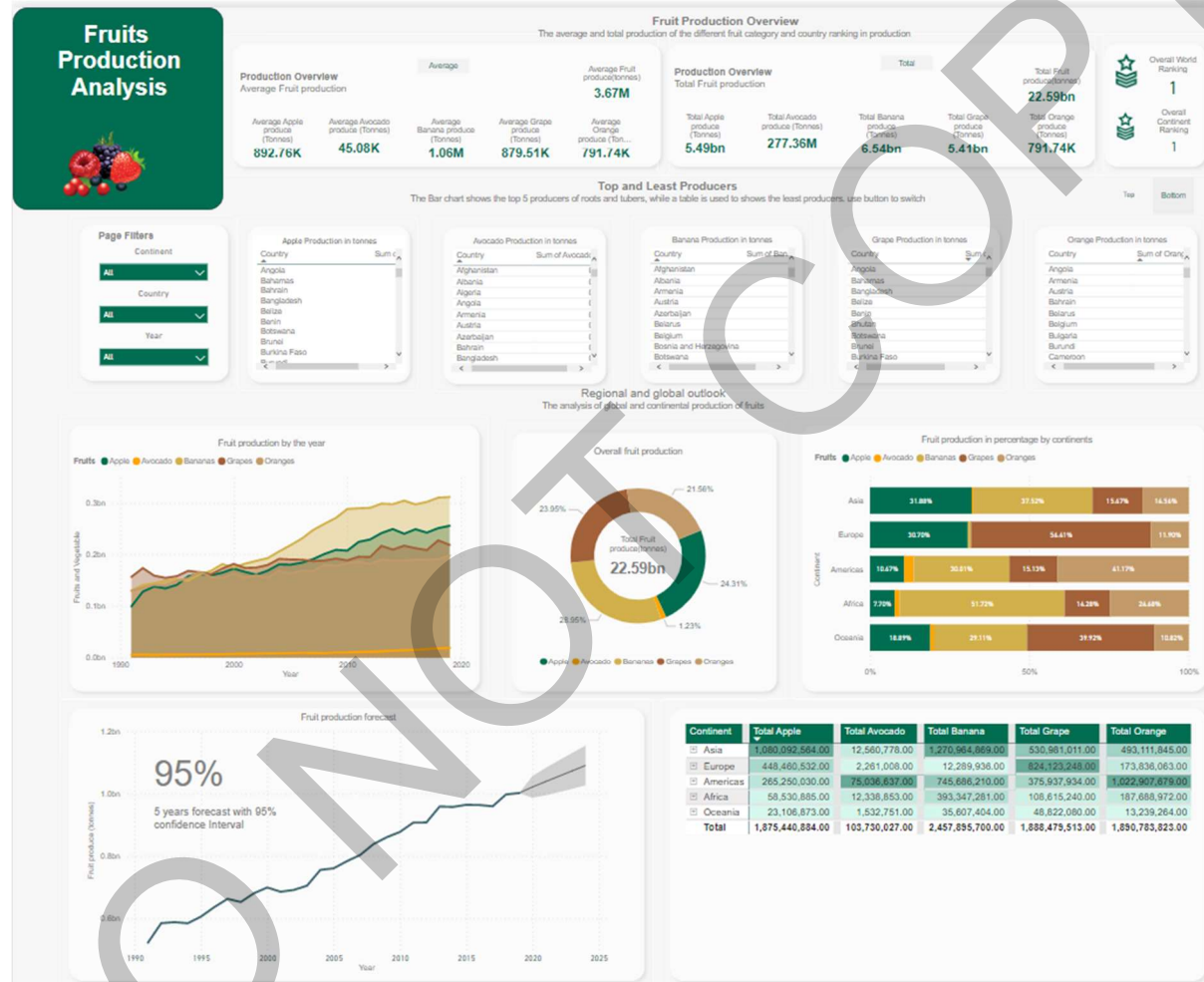
This M code is adding a new column to a table called "Renamed Columns2". The new column is called "Income category" and it is being populated with a value based on the value in the "GNI per capita" column.

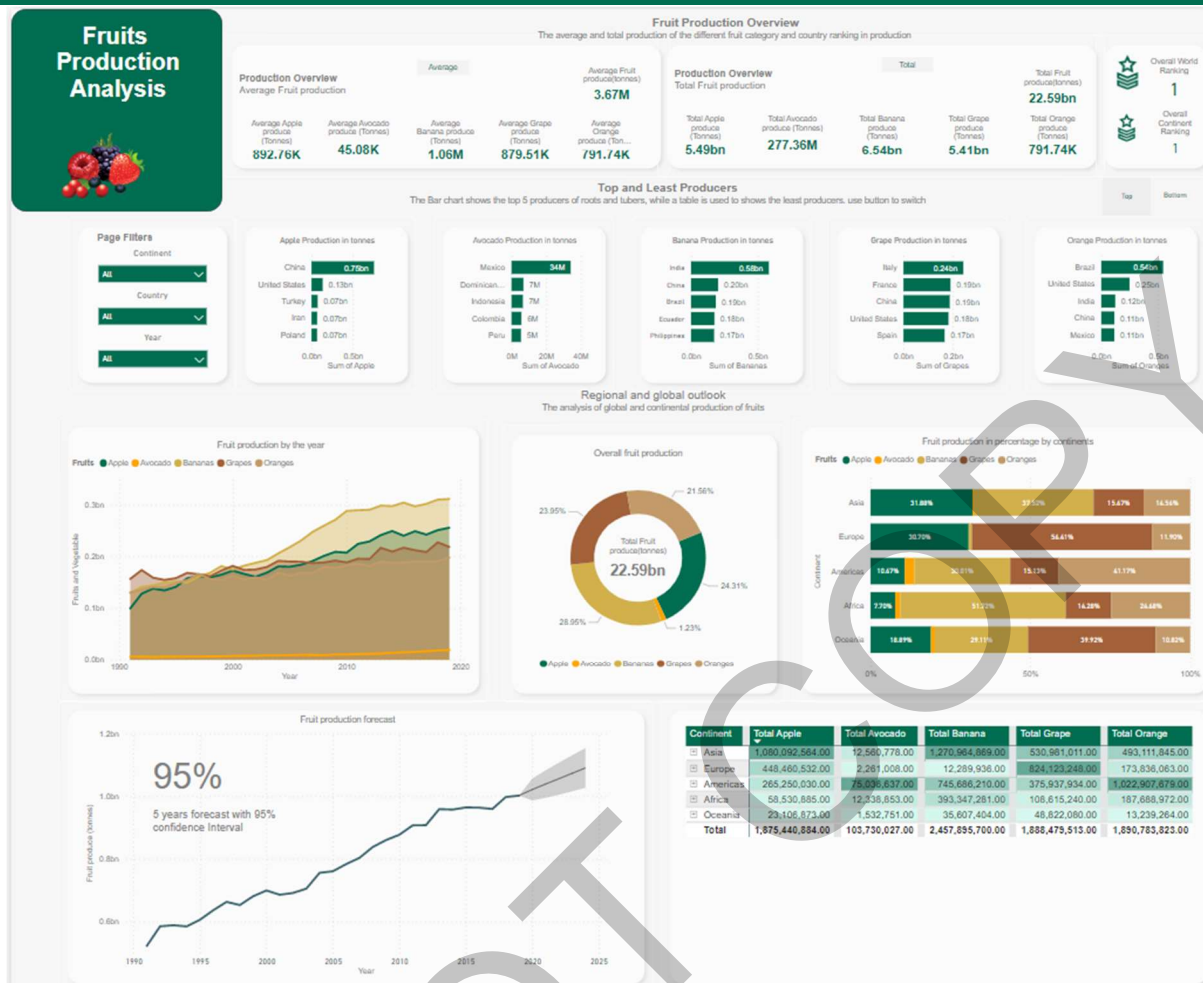
The value for each row in the "Income category" column is determined by evaluating a series of if statements. If the value in the "GNI per capita" column is greater than or equal to 0 and less than or equal to 1046, then the value in the "Income category" column for that row will be "Low". If the value in the "GNI per capita" column is greater than 1046 and less than or equal to 4095, then the value in the "Income category" column for that row will be "Lower-Middle". If the value in the "GNI per capita" column is greater than 4095 and less than or equal to 12695, then the value in the "Income category" column for that row will be "Upper-Middle". If the value in the "GNI per capita" column is greater than 12695, then the value in the "Income category" column for that row will be "High". If none of these conditions are met, then the value in the "Income category" column for that row will be an empty string.

4) Bookmarks

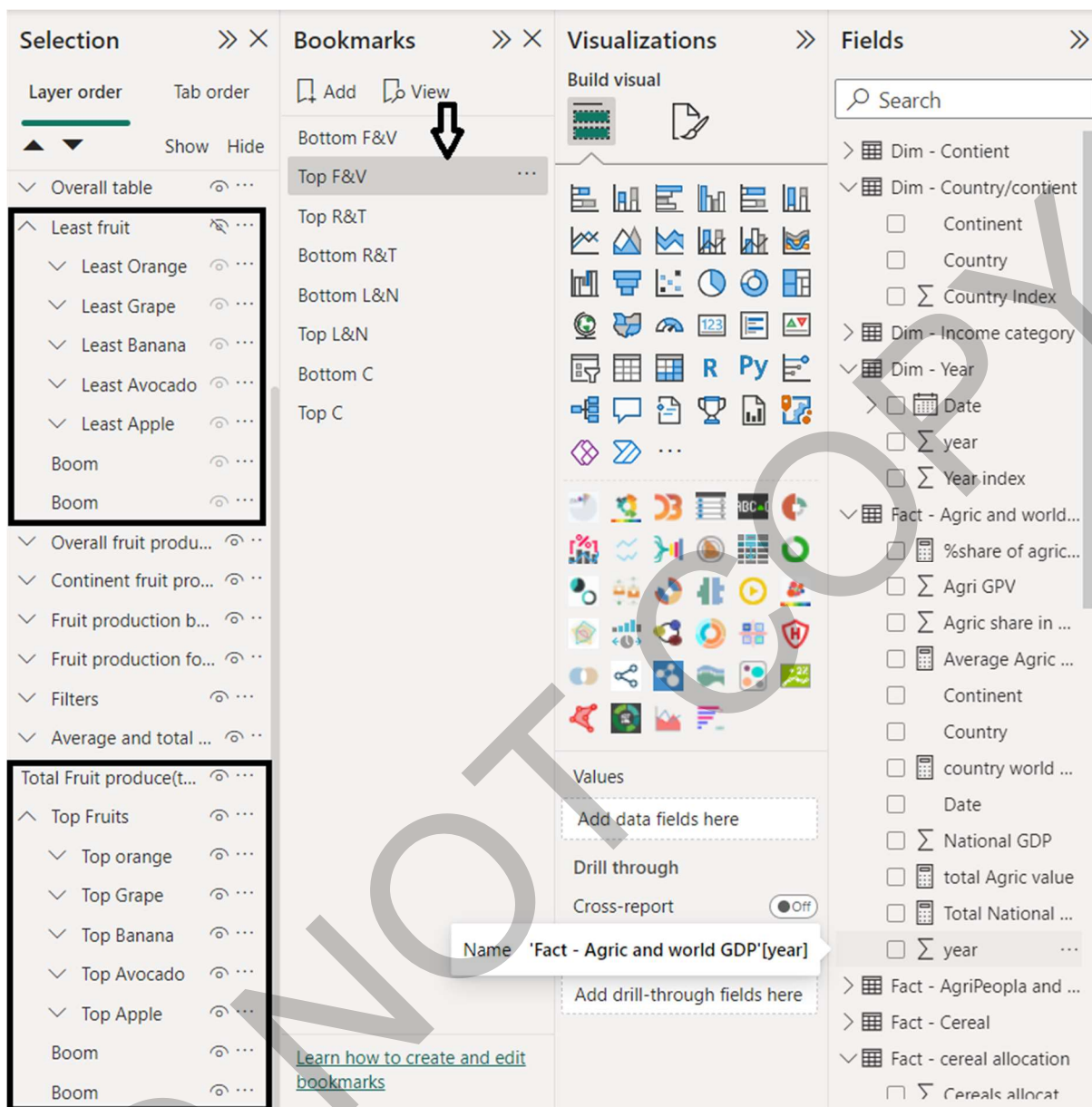
Because of the wide range of options they offer, bookmarks have proven to be extremely important in dashboard designs. However, in this report, bookmarks were helpful in combining two charts onto one page to manage space. They were also helpful in navigating to different pages from the home page.

In the fruit, cereal, legume & nut, and the Roots & tuber pages of the dashboard, Bookmarks were useful in helping the display of top and bottom producers of the various food crop types using predefined buttons and icons. The methodology used in this implementation is discussed below.

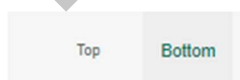




Firstly, the charts of the top producers were produced, and a bookmark of the page is then recorded. On top of each created top charts a table is then placed such that it overlaps the top chart. The table holds the list of least or non-producers of the crop type. The chart displaying the top producers and those displaying the least producers were grouped in the selection page to avoid confusion.

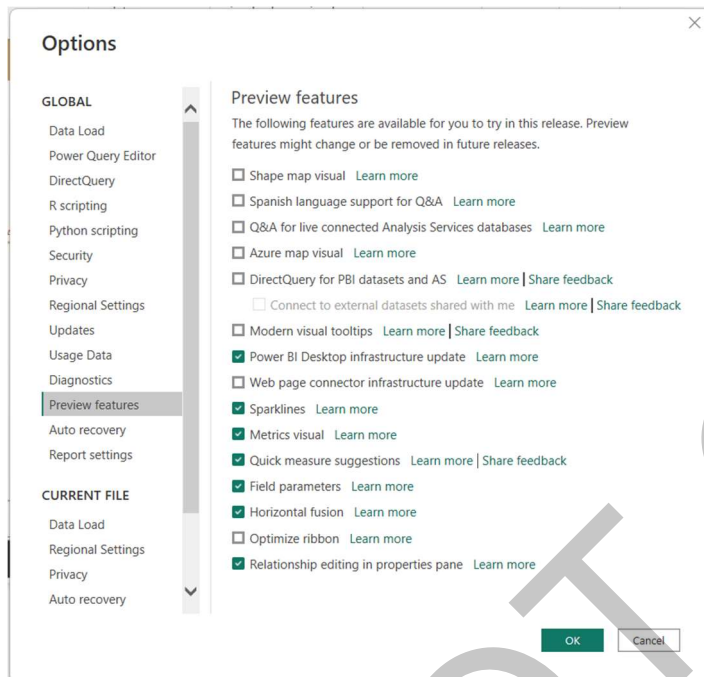


Notice the how the bottom charts were unselected in the Top F&V bookmark so that only the top charts are displayed. This step is also performed for the bottom chart and the bookmark were then linked to the Top and Bottom button respectively as shown below. The boom are the different buttons associated with the Top and bottom bookmarks. Two buttons of top and bottom are placed such that they overlap each other. In the bottom bookmark, the bottom text on the bottom button is given a green font colour and shaded grey while the top button was left unedited. This process was also repeated for the top bookmark.



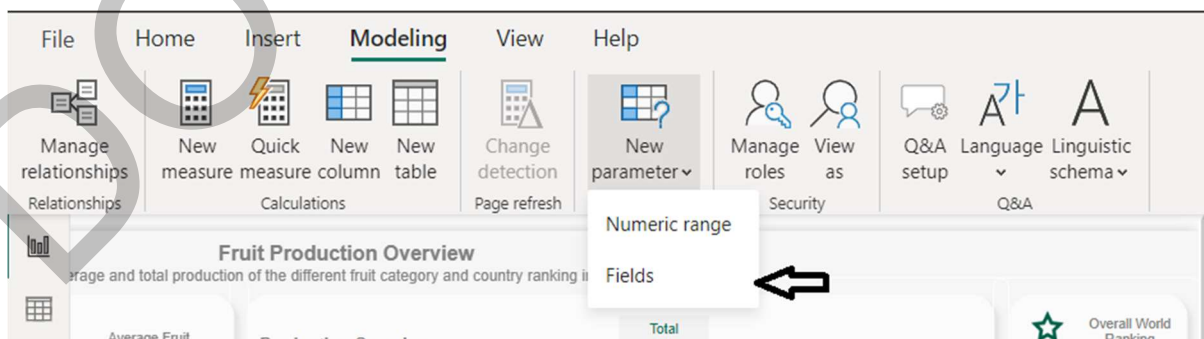
5) Field Parameters

Users can dynamically alter the metrics or dimensions being analysed inside a report by using field parameters. By choosing the various metrics or dimensions they are interested in, this feature can assist your report readers in exploring and personalising the study of the report. The Field parameter can be activated by ticking the field parameters option in the option menu of the option and settings window.



Field parameters can be used to dynamically change the parameters of a chart dynamically. However, the only constraint is that it works with measures and categorical variables majorly. In this report, a separate page has been created to enable the user of the dashboard explore for themselves by providing them with the measure of all features used in the report. The steps taken to implement this process is shown below.

After the activation step stated above, clicking the field parameter option from the modelling tab opens a window that allows you to select your desired field parameters.



Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and see different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

X axis

Add and reorder fields

- Continent
- Country
- Year

☒ Add slicer to this page

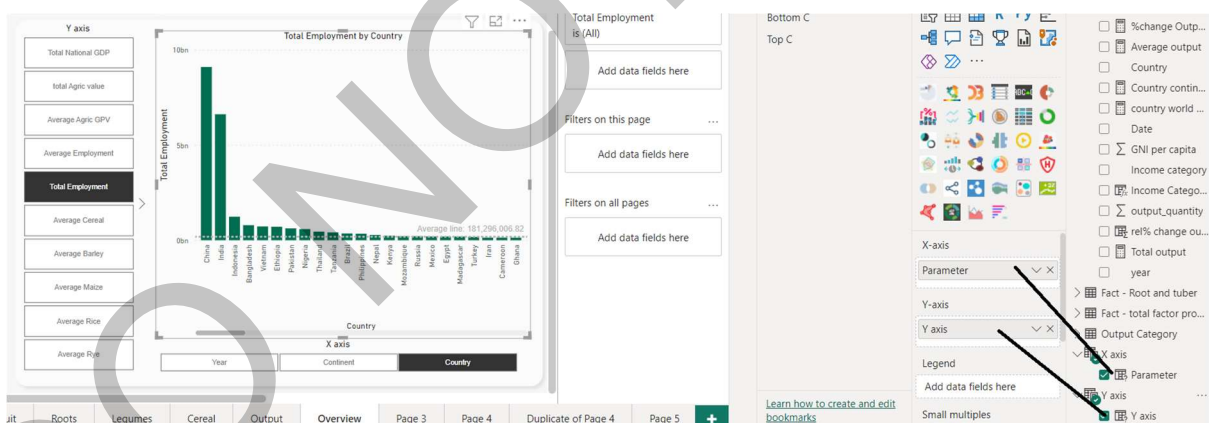
Fields

Search

- ☐ Continent
- ☒ Country
- ☐ Country Index
- ☒ Dim - Year
- ☐ Date
- ☒ Date Hierarchy
 - ☒ Year
 - ☐ Quarter
 - ☐ Month
 - ☐ Day
- ☐ year
- ☐ Year index

Create Cancel

After the desired selection has been made, ticking the add slicer option automatically displays the selected parameter on the dashboard page. On clicking the create option, a new table holding the various selected parameters will be created. However, it does not work until it has been assigned to a particular field. An example is shown below.



The X axis parameter is placed on the X axis field of the column chart and the Y axis parameter on the Y axis field. On both side of the column chart are the field parameters used for the X and Y axis of the chart.

6) Animation

The animated chart in the Animation and story page of the project was done following the steps discussed below.

- The first step involves getting the unique year values from the fact table.

```
Selected year = DISTINCT('Dim - Year'[Date].[Year])
```

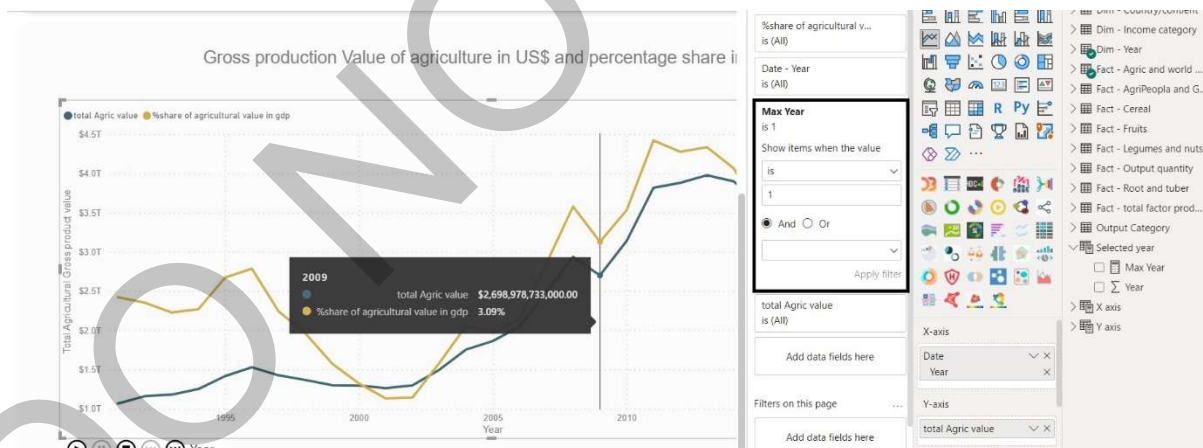
The function above creates a new table 'selected year' with the unique values of the year dimension table.

- The second is to create a calculated measure to return 1 if the year from the year column in the year dimension table is less than or equal to the year in the new table 'selected year' or 0 if otherwise.

```
Max Year = IF(MAX('Dim - Year'[Date].[Year]) <= MAX('Selected year'[Year]), 1, 0)
```

Importantly, the new table is not connected to the data model, it is left independently.

- Finally, the year column in the 'selected year' table is then placed as the parameter of the play axis and the 'Max year' measure calculated is then added to the filter of the main chart 'The line graph' and set to 'is' 1. This enables the visual to filter and display the different values for the years one after the other, in an incremental manner.



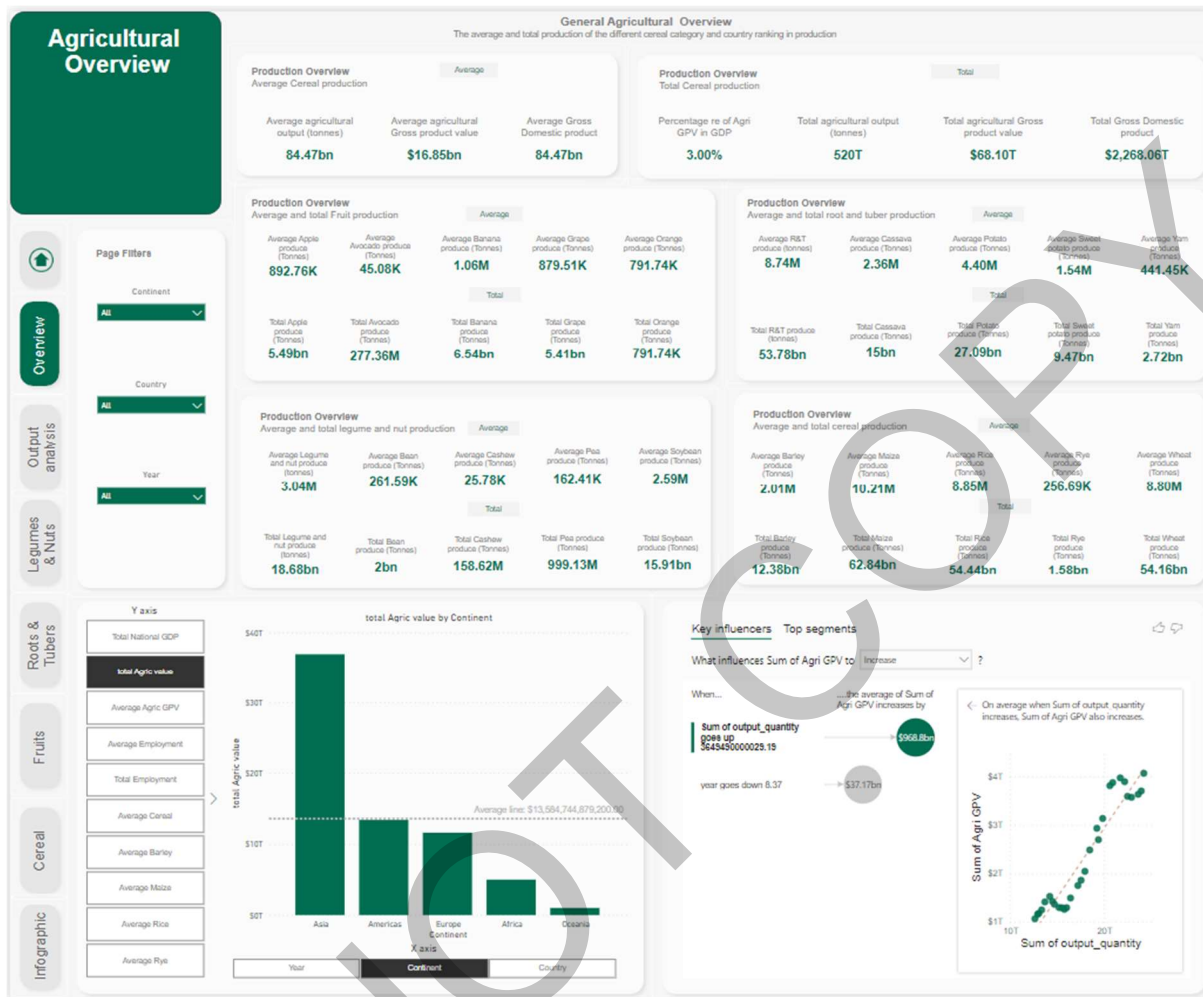
7) Dashboard

Home page



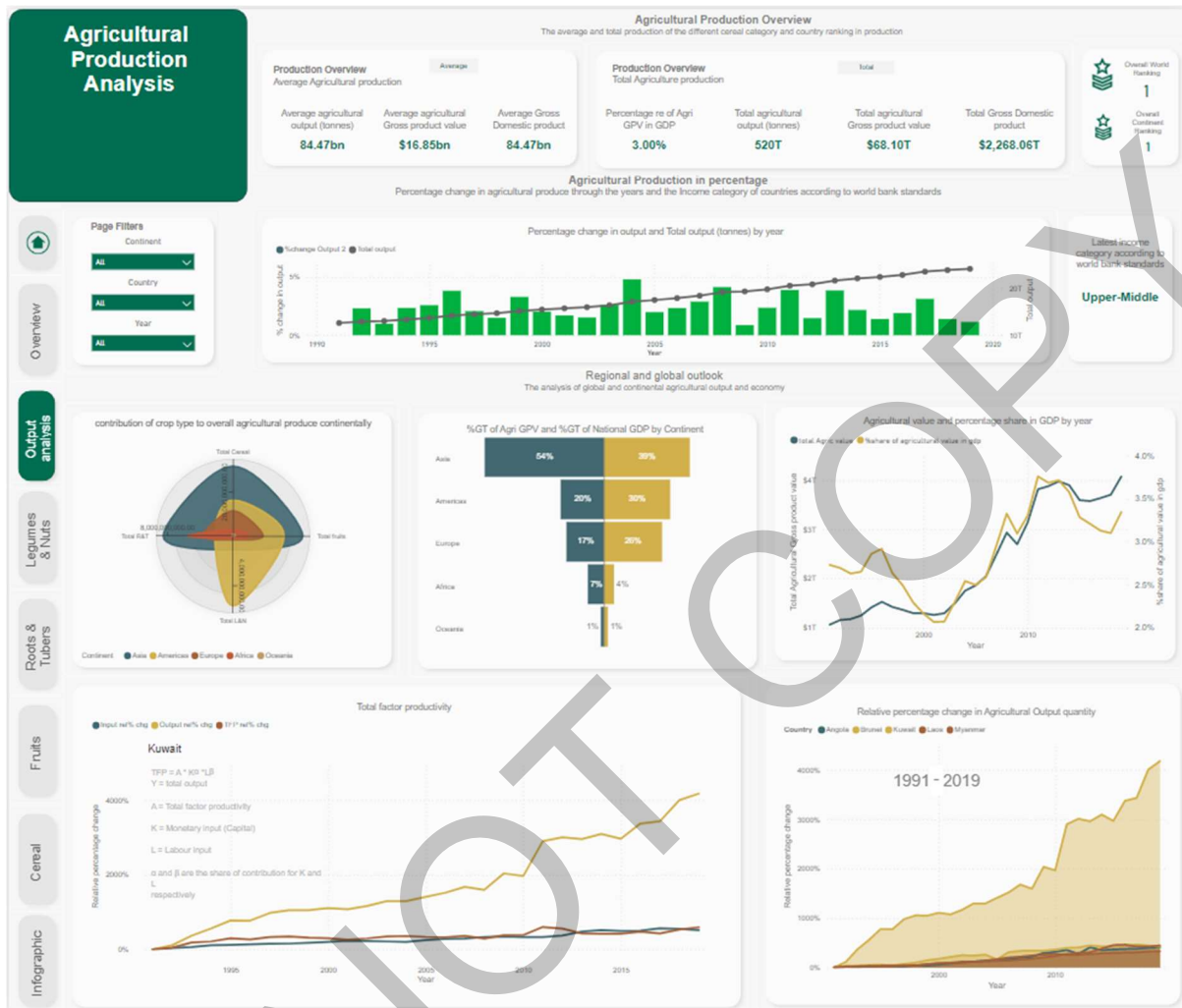
The homepage contains the general overview of the different pages in the dashboard as well as a link for accessing them.

Overview page



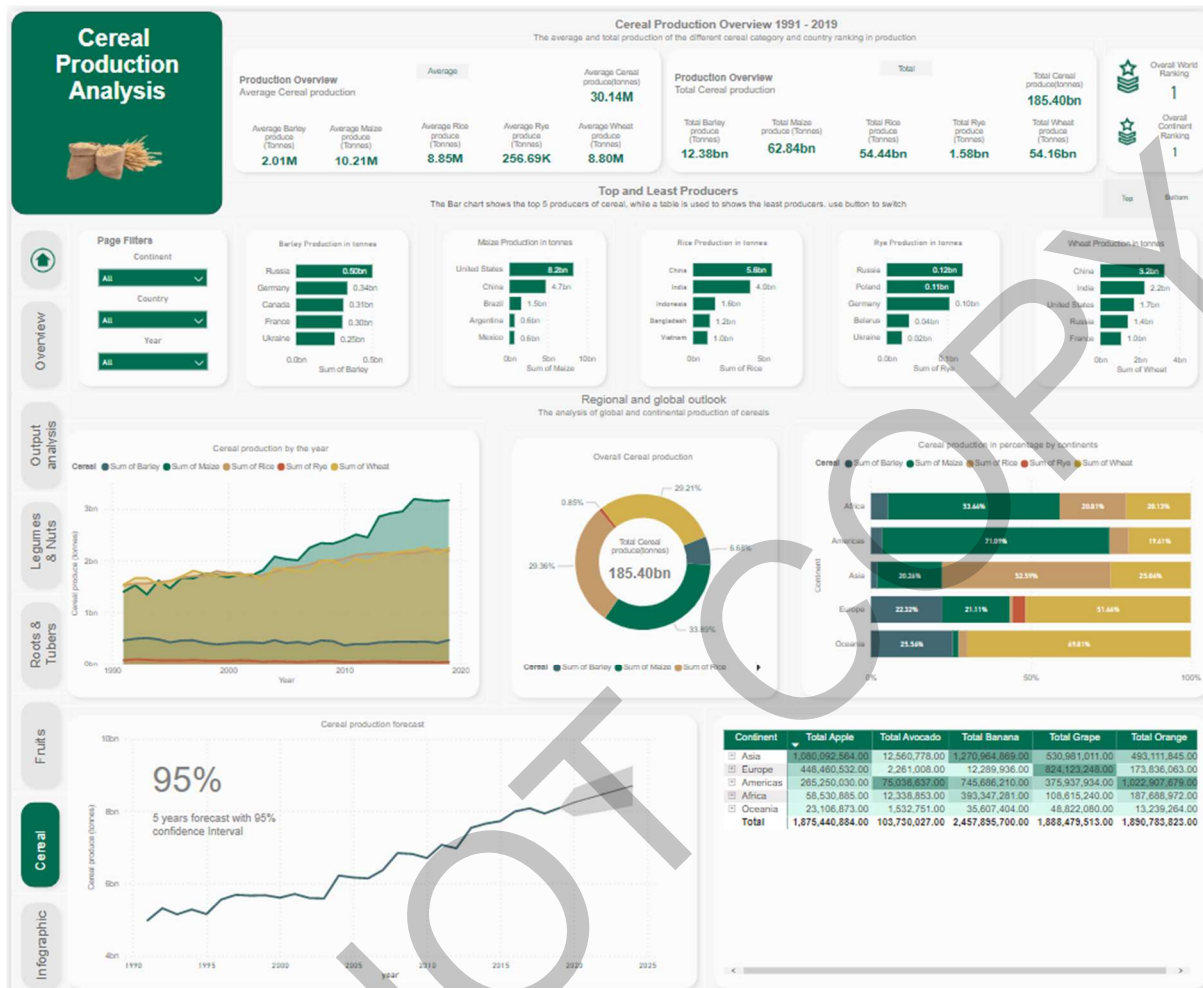
The overview page contains almost all relevant information in the analysis. Most importantly, the field parameter visual below, gives the user leverage to discover insights for themselves by providing various DAX calculated fields and categorical columns called the Y axis and X axis respectively.

Agricultural production page



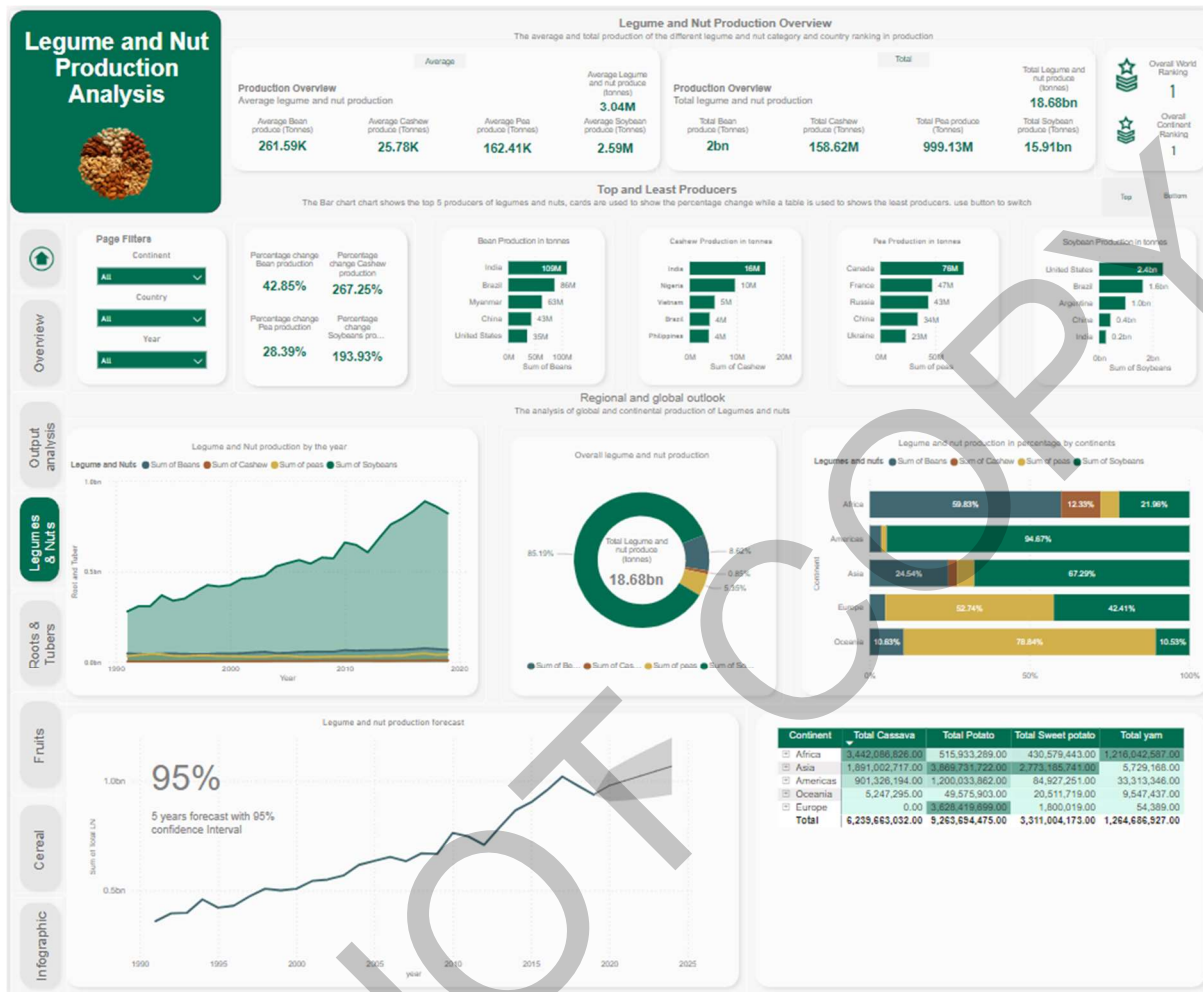
The agricultural production analysis page summarizes the agricultural produce of the various crop type (cereal, fruits, legumes, nuts, roots and tuber), the percentage contribution of agricultural production value to GDP, total factor productivity and the relative percentage change in agricultural output from 1991-2019.

Cereal production page



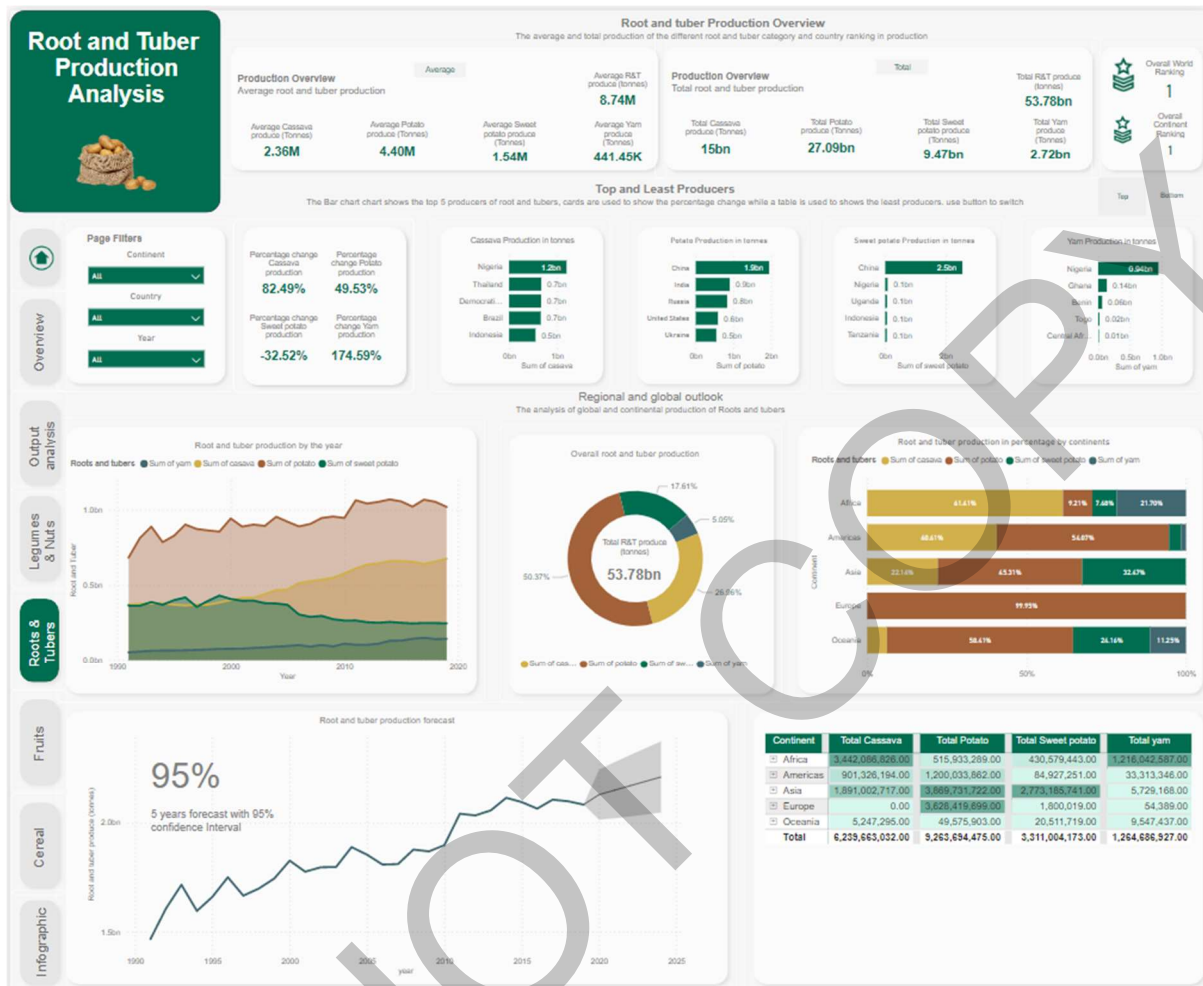
The cereal production page presents a global outlook into the production of cereal types, overall cereal production through the years, percentage proportion of cereal types in overall cereal production, continental analysis of cereal production and the forecast of future values of cereal produce in tonnes.

Legume and nut production page



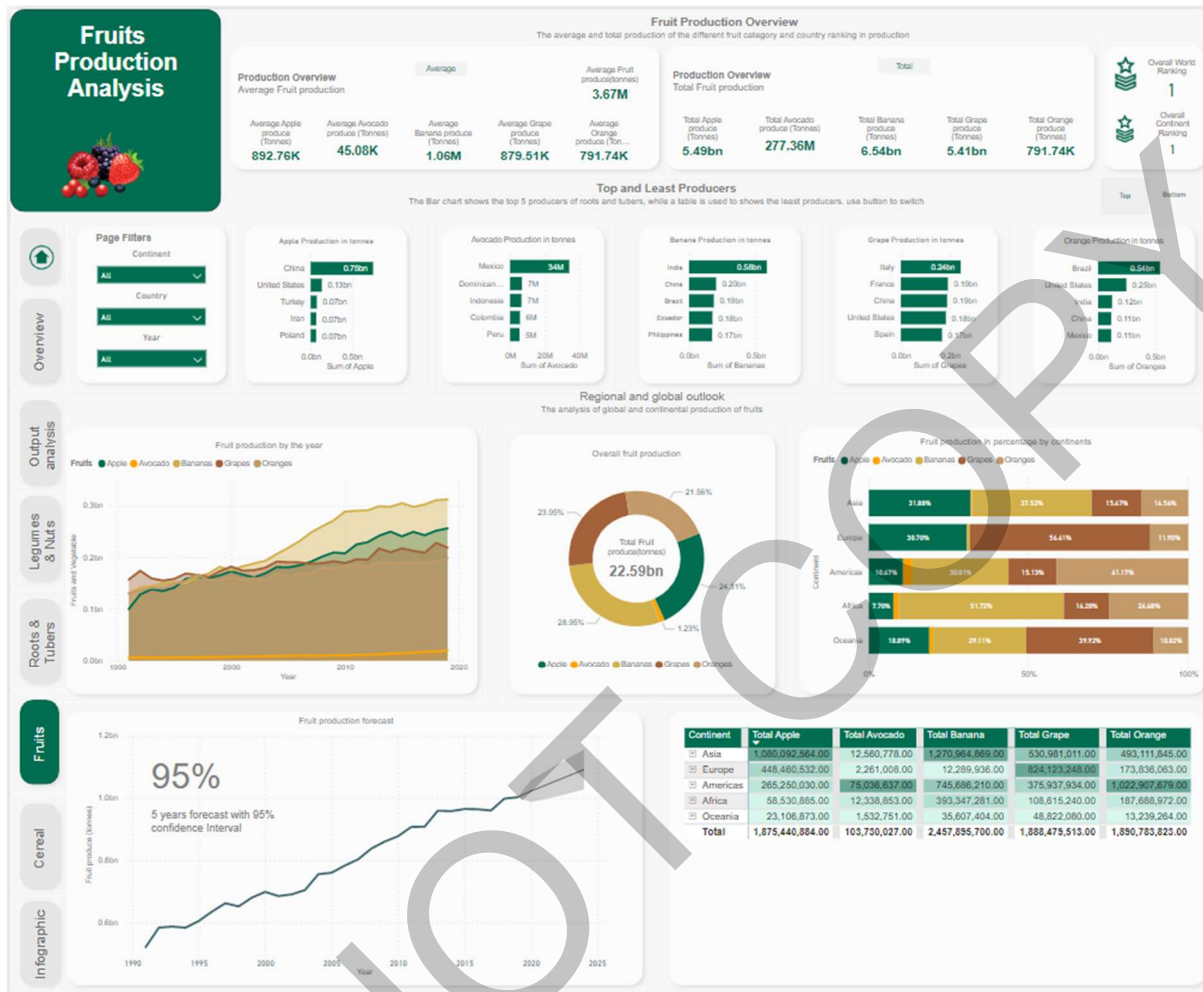
The legumes and nuts production page presents a global outlook into the production of legumes and nut categories, overall production through the years, percentage proportion of legumes and nut types in overall production, continental analysis of production and the forecast of future values of legumes and nut produce in tonnes.

Root and tuber production page



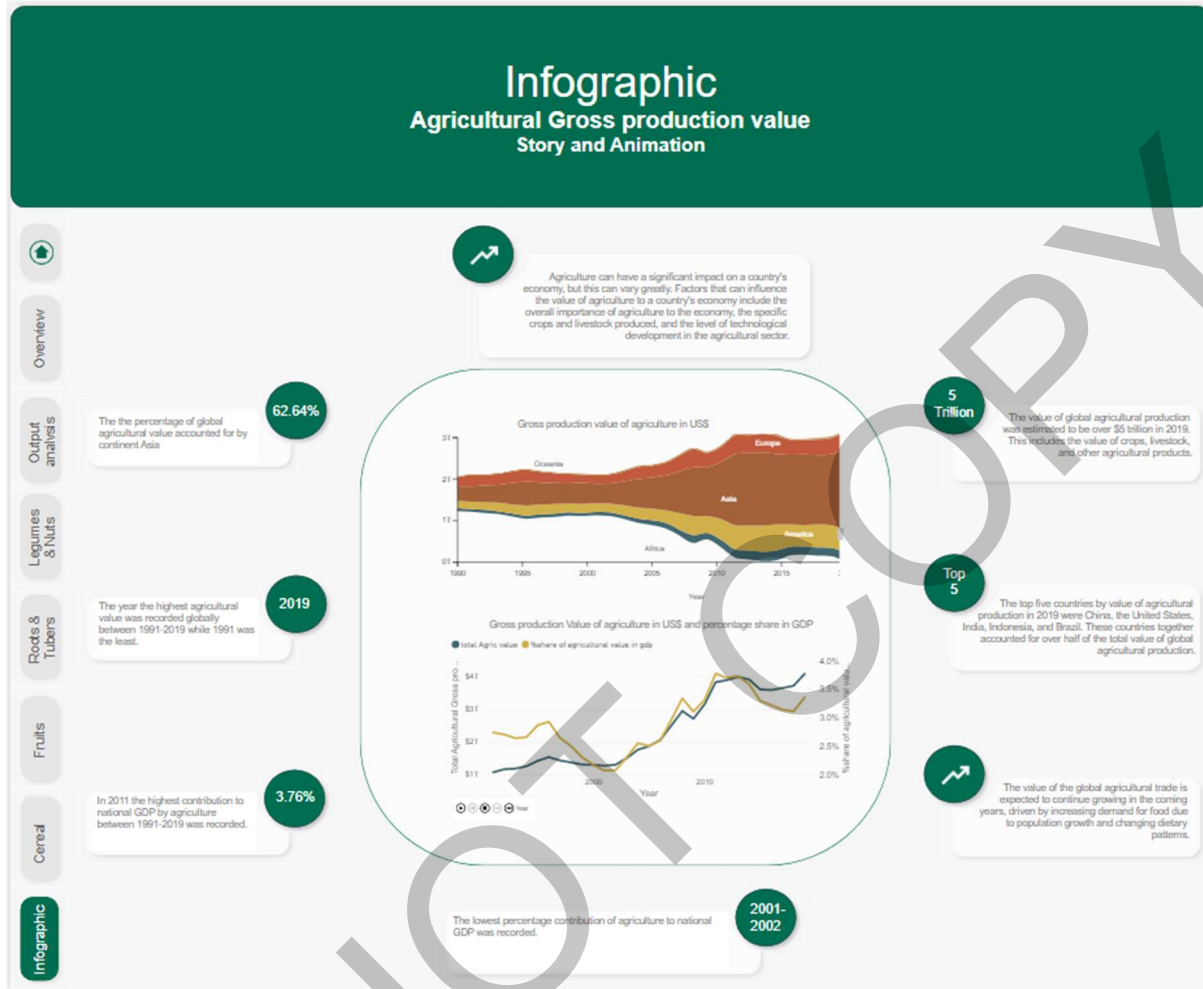
The root and tuber production page presents a global outlook into the production of roots and tuber crops types, overall production through the years, percentage proportion of root and tuber types in overall production, continental analysis of production and the forecast of future values of roots and tuber produce in tonnes.

Fruit production page



The fruit production page presents a global outlook into the production of fruit types, overall production through the years, percentage proportion of the different fruit types in overall production, continental analysis of production and the forecast of future values of fruit produce in tonnes.

Infographic page



The story and animation page presents a deeper look into agricultural gross production value from 1991-2019 and some interesting facts.

Report Section	Description	Grade
Report Structure	The creation of this report outlining all the highlighted sections in the template was led by a professional standard.	95
Data pre-processing and data modelling	The project example made use of additional processing stages that weren't addressed in the module, like R scripting. The type of the dataset used meant that the model that was chosen also required more investigation. A suitable data model, however, was developed to guarantee that no data was lost. The country and year might have been used to simply join all the tables; however this was not the best approach because data were lost. To solve this, we combined all the distinct crop types that fall under a certain crop group, for instance, cassava, yam, potato as root, and tuber table. This ensured that little or no data was lost in the process.	93
Dax and M language	Both Dax and M language has been used extensively in this report. Some examples include using Advanced functions like EARLIER, ISINSCOPE, RANKX etc. Which if not used properly will lead to data misrepresentation. Moreover, in depth explanations were provided for each DAX and M language code used to guide any user into its usage.	95
Dashboard Design	The dashboard was more than just a display of visuals. The dashboard was made with professionalism at the core.	97
Average		95